

RELEASE HISTORY

Changelog

Versioned features, fixes, security changes, and documentation updates.

2026 EDITION / IAIN BENNETT / MIT LICENSE

This file records user-visible VirtualGlove changes. The project follows Keep a Changelog categories and uses Semantic Versioning. Git remains the authoritative record for line-level and file-level history.

Stable-version entries focus on what a player, installer, or cabinet owner will notice. Release-candidate entries keep a little more workshop detail for readers who enjoy seeing how the machine was tuned, but they explain the practical result before naming the underlying mechanism.

[Unreleased]

[0.7.0] - In preparation

Added

- Bundled Controller Router 0.3.0 with explicitly registered native RetroPie adapters, merged input, Daphne exit handling and the expanded Players and Systems catalogue.
- Bundled independent UNO Q runtime 0.3.2. Older product bundles carrying runtime 0.3.1 cannot replace it during later installation or repair.

Changed

- The entry page now recovers in the background without restarting Docker, product services or the Matrix app. Healthy entry containers remain running.
- Refreshed shared Router guides and the Installation Guide filename. Installer package checks require the new shared modules.
- Restored Python 3.7-compatible recovery tests and expanded shared Router CI. Full UNO Q host and wheel support remains distinct from scoped Python 3.7 console support.

Release status

This release is prepared but unpublished. Published downloads remain on v0.6.0; the website uses `published_release_tag` until final publication. Set that field to v0.7.0 only when preparing the final publication revision and its website assets.

The prior v0.4.2 upgrade acceptance is retained as historical evidence. Candidate 0.7.0 installation, upgrade from 0.6.0, rollback, and both product installation orders still require staged acceptance. Earlier gameplay and UNO Q recovery checks apply to their recorded development builds. Preserve pairing, saved players, camera choices, game registrations, settings and controls. This release does not include separate uncommitted development work.

[0.6.0] - 2026-09-30

Added

- Added shared Controller Router pairing, the Apps chooser, and automatic app selection when a registered game starts. VirtualGlove and R.O.B. Vision can remain available together while one app controls the live game.
- Added Players and Systems settings. Fresh installations use Router for NES; upgrades retain existing routing choices.
- Added Recalbox integration and shared Matrix ownership through Controller Router. The Matrix returns to its neutral display when no game is active.

Changed

- RetroArch routing now resolves merged controllers by identity for each launch, with a legacy path for older builds and native reservations where supported. Runtime routing no longer rewrites saved RetroArch configuration.
- Camera startup restores automatic exposure when that behaviour is selected. Secure shared Setup provisions separate app access from one confirmed console connection.
- Updated Help, installation and technical guides, and the bundled Controller Router service.

Fixed

- RetroPie installation and Super Glove Ball core selection now keep the NES `retroarch.cfg` and VirtualGlove-managed `emulators.cfg` files owned by `pi:pi`, including when an earlier sudo-run update left them owned by root.
- Removed the remaining `APP_HOME` warnings when App Lab starts or stops the UNO Q application without forwarding its environment to Compose. The included services now default to the standard VirtualGlove installation directory while still honouring an explicit `APP_HOME` for development.

[0.5.3] - 2026-09-26

Changed

- Reorganised the Controller Setup page into clearer, guided sections without changing pairing or Controller Router behaviour.
- Made the standalone UNO Q enclosure more compact while keeping its screen visible and the external USB-C hub connection accessible. Updated the print files, enclosure guides, and fit guidance to match.
- Added explanatory comments around important runtime paths and clarified how to recognise and troubleshoot delayed input on Batocera and other consoles.

Fixed

- The Controller website is available at its local hostname without `:8088` when port 80 is free, and VirtualGlove releases that port when stopped. This does not take port 80 from another application.
- Suppressed an unnecessary `APP_HOME` warning during VirtualGlove shutdown and installation.

[0.5.2] - 2026-09-22

Fixed

- Fresh Recalbox installs now give the selected physical joypad working exit, menu, save, and load hotkeys from the first game, then follow Recalbox's generated bindings when they become available. Pairing on supported consoles also waits for a receiver restart to finish before reporting a failure.

[0.5.1] - 2026-09-21

Changed

- Released the validated 0.5.1 candidate as stable. Batocera's smoother Wi-Fi response, Controller Dock V2.1, and release-package checksum checks are now available through the normal install-or-upgrade commands.
- Added an Engineering page to the independently published website. It explains which technical guide to read for system architecture, input modes, native emulation, measurement, configuration, and project history.
- Clarified that enclosure fit coupons are small test prints, not pieces to install in the finished case.

[0.5.1-rc.1] - 2026-09-21

Added

- Added read-only Batocera and Recalbox FCEUmm latency preflights and a matched physical-controller/VirtualGlove acceptance record. The checks distinguish Controller timing, console input-chain health, and measured screen response without claiming an unmeasured receiver or network delay.
- Added the photo-validated Controller Dock V2.1, which replaces the withdrawn V2 design while retaining the compact UNO Q Case and Controller Dock V1. The revised enclosed dock has a wider cable-loop bay, low strain-relief guides, slimmer UNO Q posts, more vertical plug clearance, a shorter lid skirt with larger boss reliefs, a roomier wordmark recess, and inexpensive mount and logo fit coupons.

Fixed

- Release assets are now uploaded as a draft and downloaded again for checksum verification before the candidate is published. Batocera's installer still refuses a mismatched ZIP without changing the installed system.
- Batocera now disables power saving on a connected Wi-Fi interface when VirtualGlove starts and before each game. This prevents wireless buffering from delivering otherwise intact hand-position updates in jerky bursts; wired connections and disconnected Wi-Fi interfaces are untouched.
- Moved the V2.1 hub cradle and Ethernet opening 14 mm forward from the rear wall, keeping the hub and its cradle clear of the rear-right cover screw tower while retaining rear cable access.
- Removed the superseded V2 STLs, bundle, renders, and current guide links so builders cannot accidentally choose the enclosure that failed the physical cable, board-mount, logo, and lid-clearance checks.

[0.5.0] - 2026-09-20

Changed

- Standardised public application copy, Help, the website, and maintained documentation on Canadian English, including the **Centre hand** control. Technical identifiers, filenames, protocol fields, and third-party names remain unchanged, and automated checks prevent American spellings from returning to published prose.
- Rebuilt the public website for 0.5.0 as four tracked, static pages covering the project, installation and upgrades, enclosure choices, and the creator's motivation. Release-sensitive commands and links now come from one checked facts record, and the build produces a verified ZIP for manual upload. That independently published website ZIP is no longer attached to application release candidates or releases.
- Made every platform installation example begin in a known writable folder, including an explicit extracted-package `Set-Location` step for LaunchBox. Fresh installation and v0.4.2 upgrade checks are now separate and copyable. The project README now uses the same platform commands, accounts, prerequisites, install-or-upgrade wording, and first-game checks as the full Installation Guide.
- Organised the printable enclosure files through one checked manifest and added complete UNO Q Case, Dock V1, and Dock V2 download bundles without breaking the existing individual STL and 3MF links.
- Corrected the full-wordmark Dock V1 and Dock V2 lid recesses to accept the actual 100 x 30 mm multicolour logo with printable PLA clearance. The smaller UNO Q Case continues to use its matching compact wordmark.
- Showed the receiver UDP port with the other Connection and startup settings instead of hiding it under Advanced connection.
- Console upgrades now retain the newest five routine rollback sets while preserving every explicitly named migration or recovery backup. This keeps repeated RetroPie, Recalbox, and Batocera updates from quietly consuming storage.
- Refocused the optional Engineering Toolkit around maintained workflows for guided recognition capture, labelled replay, movement response, latency, camera delivery, native emulation, and isolated accelerator research. The guide now starts with questions people are likely to investigate and points to the smallest suitable workflow.
- Added platform-aware input diagnostics to the Engineering Toolkit guide. It now covers Controller Router checks on RetroPie, Recalbox, and Batocera, LaunchBox's separate loopback RetroPad path, and the RetroPie-only boundary of the existing end-to-end preflight.
- Reduced the Engineering Toolkit source archive by excluding packaged Batocera, Recalbox, and Windows emulator binaries and their platform source archives. Native researchers still receive the editable VirtualGlove patch, diagnostic header, and safe local builders.
- Kept native-core matrix builders, documentation renderers, the code-review map generator, and the test runner in full development checkouts instead of copying maintainer-only commands onto ordinary Controllers and consoles.

Removed

- Removed three superseded engineering utilities: the timer-driven camera recorder, the retired `min/boost` trace-name comparator, and the earlier ideal-step motion sampler. Guided recording, replay evidence, and the bounded native-motion curve are their maintained replacements.

Fixed

- Kept Batocera's merged physical controller active in every Libretro system, not only NES. Controller Router now writes a bounded, reversible `batocera.conf` override that Batocera's own generator applies after choosing frontend controllers, so Game Boy, SNES, Game Gear, and other Libretro cores no longer open the physical device that Router has exclusively grabbed.
- Corrected the printable Controller enclosures after a physical fit check. Narrower UNO Q supports now clear its underside connectors, lid skirts pass around the corner bosses and thick USB-C plugs, and Dock V2 provides a wider left-side bend area for the hub's captive cable while keeping Ethernet aligned with its side opening.
- Made enclosure exports reliable on the affected Apple Silicon OpenSCAD build by using its working headless x86-64 application slice when the arm64 Qt startup incorrectly rejects NEON support.
- Kept Dashboard controls recoverable when a profile change is accepted but never confirmed, limited broken camera-stream and registry retries, and kept the real registry error visible instead of replacing it with "game not found."
- Made Glove Academy genuinely retry a failed practice-camera lease instead of only saying it would retry, and ensured leaving the page always sends a safe release even when startup was interrupted.
- Restored the lesson view cleanly when Tune gestures cannot start or stop, so a failed personalisation lease cannot leave its switch off with the tuning panel still covering the Academy.
- Removed the confusing Get ready to play experiment. Setup, Dashboard, and Glove Academy now remain the direct paths for connection, centring, practice, and starting controller output; upgrades discard only its isolated progress.
- Kept Connection and Camera saves independent. Saving one section no longer applies or discards unsaved edits in the other, and camera-only edits no longer block console pairing.
- Removed a retired native-motion command-line option from the full camera exposure soak and strengthened the toolkit self-check and package tests so duplicate inventory entries or compiled emulator artifacts cannot quietly return.

[0.5.0-rc.1] - 2026-09-20

VirtualGlove 0.5.0 expands beyond RetroPie, adds Windows support, and completes the move from the old development name to VirtualGlove. The emphasis is on a predictable installation: choose the platform, pair once, keep the physical controller available, and let each game use the correct input method.

Fixed

- Restored the release checks on every supported Python version and installed the PDF preview tool in the automated packaging environment. Release builds now exercise the same enclosure previews and compatibility paths as local validation before a candidate can be published.
- Completed the LaunchBox in-place namespace upgrade. Installations that used the current package name with the retired Python module folder now remove only that leftover managed runtime folder and command files, while preserving pairing, game registration, emulator settings, ROMs, saves, and controllers.
- Followed controller remaps made in EmulationStation automatically. Controller Router keeps the selected Player 1-4 assignment, refreshes the validated button mapping while idle or before the next game, and lists newly configured controllers as unassigned. An incomplete mapping now disables only that source instead of risking scrambled controls or hotkeys.
- Started every routed NES game from a neutral VirtualGlove state. Physical controllers now work immediately at launch, while gestures are admitted only after one fresh neutral hand observation. The joystick route no longer forwards native camera-position axes, preventing a frontend gesture or off-centre hand from blocking the title screen.
- Kept cabinet and joypad controls responsive while VirtualGlove is active. Controller Router now services physical input first, discards queued camera history in bounded batches, and avoids rescanning every input device on each control cycle. It also preserves standard Home/Guide hotkeys and the arcade cabinet's established I-PAC hotkey without allowing VirtualGlove or Players 2-4 to trigger RetroArch shortcuts.
- Kept RetroPie's Controller Router socket available when the receiver restarts. The receiver no longer removes the Router-owned runtime directory, upgrades restart both services in dependency order, and installation checks now catch a missing Router socket before gameplay.
- Matched Controller Router's player indexes to each platform's RetroArch generation. RetroPie now follows its stable `jsN` assignment while Recalbox and Batocera retain their udev joypad ordering. RetroPie's root-owned Router service keeps these core assignments current, and its unprivileged launch hook copies the validated block into RetroPie's final temporary launch file without rewriting protected persistent configuration.
- Extended Controller Router to stock Nestopia as well as FCEUmm, while keeping Nestopia (VirtualGlove) on its separate native-input route. Recalbox and Batocera can retain their chosen ordinary NES core without losing either the physical controller or VirtualGlove.
- Applied routed physical-player assignments across Libretro systems while keeping VirtualGlove gestures limited to supported NES paths. Recalbox now stores the managed assignment in its persistent ROM-root override instead of the launch-generated temporary file. Physical controller acceptance passed in Game Boy, ColecoVision, and Game Gear as well as NES and native Super Glove Ball.
- Added a red Controller Router warning when no physical controller is assigned to Player 1, because RetroArch menu and exit hotkeys may then be unavailable during NES joystick gameplay.
- Improved Controller Router table spacing with clearer status and player assignment headings, balanced column widths, and readable small-screen scrolling.
- Gave Controller Router its own Setup card between Connection and startup and Pair this Controller, keeping controller assignments separate from pairing.
- Made the Controller Router wait for Setup to restore the saved console platform before checking assignments. An already paired console is no longer briefly or permanently described as needing pairing again.
- Simplified the Setup dead-zone camera test by hiding redundant direction labels and idle status text. Its explanation now appears only while the camera test is on.

- Kept multi-player USB boards such as the I-PAC Ultimate I/O as separate logical controllers. Controller Router now distinguishes the board's Player 1 and Player 2 interfaces while still recognising the same board after a USB port move.
- Fixed Recalbox and Batocera merged Player 1 assignment when Linux's `jsN` number differs from RetroArch's joypad order. Ordinary FCEUmm and stock Nestopia games now use the same merged controller that RetroArch actually sees.
- Preserved the real button codes supplied by Recalbox and Batocera controller mappings. Controllers that report Start, Select, or Home outside the legacy joystick button range now retain their correct face-button and menu layout.
- Made the merged controller the sole reader of the selected physical Player 1 pad during gameplay. The original pad remains untouched in EmulationStation, while games receive one canonical mapping instead of competing physical, keyboard-hotkey, and merged interpretations of the same press.
- Added the complete Batocera libretro hotkey layout to the merged pad and refresh its current RetroArch index immediately before each game. USB event numbers and legacy joystick numbers can now change without changing Player 1.
- Upgrading an UNO Q from 0.4.2 now removes the old background services only after their VirtualGlove replacements are running. If another boot component is already using the Matrix connection, the optional early-start feature steps aside cleanly instead of leaving a failed service warning.

Added

- Added packaged Nestopia (VirtualGlove) cores for RetroPie's ARMv6, ARMv7, 32-bit ARMv8, AArch64, and x86-64 targets. Installation chooses from the actual RetroArch executable format rather than the kernel name, verifies the binary and corresponding GPL source archive, and retains local compilation as a fallback when no packaged target matches.
- Added optional **Controller Router** for RetroPie, Recalbox, and Batocera. It can combine several EmulationStation-configured controllers into any of four FCEUmm players and assign the single paired VirtualGlove to one player. Setup provides review-before-save assignments, a short live-input check, revision-safe saves, and rollback. Standard RetroPie remains unchanged until Router is explicitly enabled.
- Migrated the VirtualGlove development cabinet from its original fixed I-PAC/8BitDo merger to Controller Router after reboot, hotkey, physical-pad, VirtualGlove, and simultaneous-input acceptance. The original merger remains available as a tested rollback reference.
- Added a console-local Controller Router assignment screen with connection status, Players 1-4, live input testing, save confirmation, and rollback. Controller assignment remains separate from secure console pairing.
- Recalbox and Batocera now migrate the released merged Player 1 record into Controller Router without changing its controller, mapping, hotkeys, or VirtualGlove assignment. The former Player 1 record remains available during the transition.
- Controller Router assignments are confined to FCEUmm's core-specific RetroArch override, keeping native Nestopia and unrelated NES cores outside the routing boundary. The recovered cabinet integration now includes its exact RetroArch profiles plus receipt-gated migration and one-command rollback tooling.
- Added an advanced per-game `four_score: "force"` option for compatible ROM variants that FCEUmm's CRC list does not recognise. Known Four Score games continue to use FCEUmm's automatic detection.
- Added printable VirtualGlove Controller enclosures: a compact UNO Q case, a new full-access Controller Dock V2, and the retained low-profile Dock V1. V2 hides the Arduino hub inside a fully closed case while rear USB-C access, internal USB-A cable routes, and a dedicated side Ethernet opening keep the required connections available. Parametric OpenSCAD sources, manifold STL files, fit-test coupons, Kobra 3 PLA guidance, and print-safe versions of the full logo and hand/target emblem are included.
- Added small-emblem and full-wordmark lid variants. Their logo backings now include locating recesses for the cyan and red inserts, and optional multicolour 3MF files keep all three materials together for an ACE workflow.
- Added first-class installation and pairing for RetroPie, Recalbox, Batocera, and LaunchBox. Setup asks for the platform before the address, then shows only the pairing instructions that apply to that system.
- Added Recalbox and Batocera support for a combined **VirtualGlove Merged Player 1** controller. Hand gestures and the chosen physical joypad appear as one game controller, while the physical hotkey still handles exit, menus, save states, and other frontend commands. EmulationStation continues using the original physical controller, avoiding doubled menu input.
- Added LaunchBox support for 64-bit Windows through RetroArch. Ordinary NES games use FCEUmm and RetroArch's built-in Network RetroPad; Super Glove Ball uses the separately named Nestopia (VirtualGlove) core. The physical XInput joypad and real keyboard remain available alongside VirtualGlove.
- Added physical Start and Select fallback for native Super Glove Ball on Windows. The joypad can handle menus while the camera supplies native hand movement and glove actions.
- Added native Super Glove Ball packages for Recalbox, Batocera, and LaunchBox. The installer chooses the correct processor build, checks that RetroArch can load it, and leaves FCEUmm available if the native core is unsuitable.

- Packaged Recalbox builds for seven supported targets and Batocera builds for fifteen. ARM32, ARM64, and x86-64 files are kept separate, and every binary includes matching source and checksum records.
- Added automatic native-core selection for registered Super Glove Ball ROMs on Recalbox and Batocera. A choice already made by the user is preserved.
- Strengthened pairing so the Controller confirms both the saved platform and address during the physical approval window. This prevents instructions for one console family from accidentally changing another.
- Improved the V-sign Start gesture for naturally spread fingers. It now looks at the straighter outer finger joints, reducing missed Start commands without changing the intentional hold, release, or Menu Guard behaviour.
- Added a reusable local test environment. Instead of reporting the same missing NumPy or OpenCV module many times, the test runner checks once and gives one clear setup command.
- Added reproducible build and verification tools for the Windows and Linux native cores. These are maintainer safeguards: an incomplete or mismatched core cannot enter a release package.

Changed

- Prepared the first 0.5.0 release candidate after physical acceptance on RetroPie, Recalbox 10.1.1, Batocera 43.1, and LaunchBox. Ordinary NES input, native Super Glove Ball, physical-controller coexistence, hotkeys, reboot persistence, and current EmulationStation mapping adoption were verified on the applicable platforms.
- Reorganized Controller Help into User Manuals, Installation Guides, Technical Documentation, and Project Information. The new VirtualGlove Input Modes guide explains joystick mode, Recalbox/Batocera merged Player 1, platform differences, and native Super Glove Ball operation in one place. Native movement results now live in the Engineering Journey; the three superseded detailed sources remain clearly marked as archival research rather than duplicate Help cards and PDFs.
- Rebuilt the Enclosure Assembly Quick Reference as an eight-page workbench manual. It now begins with a complete parts tray, then gives the UNO Q Case, Dock V1, and Dock V2 uninterrupted, numbered assembly procedures using the actual enclosure models. Pixel Pal appears only for useful warnings and the final check. The detailed Enclosure Guide uses the same step numbers while retaining print settings, downloads, fit guidance, and troubleshooting.
- Removed the pale OpenSCAD matte from the enclosure renders. Transparent source previews now sit cleanly on the Enclosure Guide, Quick Reference, and Controller Help backgrounds without yellow image boxes.
- Reworked the Architecture, Native Movement Validation, Third-party Notices, and Changelog guides so they explain the practical result before the implementation detail. Exact measurements, checksums, platform names, and licensing obligations remain available for readers who want them.
- Rewrote the Installation Guide as a concise 0.5.0 walkthrough. RetroPie, Recalbox, Batocera, and LaunchBox now follow the same install, checkpoint, first-game, and update structure; low-level build and integration details remain in the technical references instead of blocking the setup steps.
- Completed the internal rename to VirtualGlove. New installations no longer ship the retired development package or command names.
- Added a guarded upgrade from 0.4.2 to 0.5.0. It backs up files VirtualGlove owns, removes obsolete copies, and preserves players, calibration, tuning, pairing, game lists, device settings, ROMs, saves, and unrelated files.
- Linux installers now stop the running VirtualGlove input service before replacing files, preventing old and new copies from sending controls at the same time. If an upgrade fails, the installer tries to restore the previous working service and reports anything that needs attention.
- Setup no longer offers pairing-key replacement as a connection-save option. Pairing credentials change only through the clearly labelled pairing flow and still require physical confirmation on the Controller.
- LaunchBox now keeps held directions and buttons held for as long as the hand pose remains active. Returning to centre, losing tracking, closing the game, or restarting the receiver still releases every control immediately.
- LaunchBox uses one input route at a time: FCEUmm games use Network RetroPad, while native Super Glove Ball uses its Power Glove data channel. This avoids duplicate Start, Select, or action presses.
- LaunchBox starts and repairs its small background bridge whenever a managed game launches, including after a Windows reboot. Existing games assigned to another emulator are left alone, and changed LaunchBox configuration files are backed up before migration.
- Documented how to add NES ROMs after installation on RetroPie, Recalbox, Batocera, and LaunchBox, including refreshing the frontend and choosing a different emulator for an individual game.
- Setup keeps **Check console address** available during pairing confirmation and restores it after every result while connection-editing fields remain safely locked.

- LaunchBox pairing now invokes a dedicated runtime-restart wrapper so its single-use pairing command works reliably on Windows.
- Wi-Fi deployment no longer opens an invisible remote sudo prompt. It deploys and verifies the Controller application. When administrator access is needed, it prints the exact UNO Q commands for the user instead of waiting for a password in a terminal window they cannot see.
- Updated the README and every maintained installation-facing guide for the v0.5.0 platform model. Added dedicated Recalbox, Batocera, and LaunchBox setup sections, reboot checks, update instructions, Player 1 coexistence, and the safe fallback from native Super Glove Ball to FCEUmm.
- Added separate Recalbox and Batocera release packages. Installation chooses the physical Player 1 controller and preserves pairing, game registrations, ROMs, saves, controller mappings, and unrelated RetroArch settings.
- LaunchBox checks whether its optional keyboard fallback would collide with a RetroArch hotkey. A conflict disables only that fallback; Network RetroPad and the physical XInput controller continue to work.
- LaunchBox verifies the native Windows core before installation and again when Super Glove Ball starts. If the file is missing or changed, the game safely falls back to FCEUmm instead of loading an unknown DLL.
- Batocera and Recalbox now start their packaged helpers in the way each system expects, including Recalbox storage that does not retain executable file bits.
- The isolated Nestopia core selects the native Power Glove device itself, making Super Glove Ball startup less dependent on frontend timing.

Fixed

- Controller shutdown now ignores repeated stop requests while it finishes cleaning up the camera preview, avoiding a harmless but alarming traceback.
- LaunchBox FCEUmm games now use RetroArch's reliable Network RetroPad instead of simulated Windows key presses. D-pad, A, B, Start, Select, held controls, and rapid fire work while the physical joypad and keyboard remain available.
- LaunchBox upgrades now wait until the old receiver has fully stopped before replacing it, preventing duplicate background copies after an update.
- Recalbox and Batocera no longer discard valid movement or button input when a status message also contains recognition-only information.
- Merged Player 1 now carries the physical controller's RetroArch hotkeys correctly. VirtualGlove Select can never become Hotkey Enable, while the physical controller can still exit, open menus, save or load states, take screenshots, and start recordings.
- The shared Controller Dashboard now replaces an unavailable camera stream with clear USB-camera guidance instead of repeatedly retrying a broken image, consistently for RetroPie, Recalbox, Batocera, and LaunchBox consoles.

[0.4.2] - 2026-09-14

Changed

- Promoted the validated 0.4.2 release candidate to stable. Updated release labels, install commands, project status, built-in Help, and generated PDF editions to identify `v0.4.2` consistently.

[0.4.2-rc.2] - 2026-09-14

Changed

- Corrected every release-facing guide, install command, project-status note, review snapshot, and generated PDF to identify the 0.4.2 candidate. References to 0.4.1 remain only where they describe the latest stable release, historical changes, or the oldest supported in-place upgrade.

[0.4.2-rc.1] - 2026-09-14

Added

- Added per-game **Rapid A** and **Rapid B** controls to Dashboard. Changes use the paired game list and take effect immediately during play. The temporary choice ends when the game exits or a new game session begins.

Changed

- Kept frequently used device, camera, and Wi-Fi information in memory for a short time. The Dashboard no longer rereads files or rescans USB cameras on every refresh, while Setup still notices camera changes promptly.
- Compressed development deployment archives before transfer. Documentation PDF builds now downsample embedded copies to print resolution while leaving all source artwork, including Pixel Pal's intentional extra digits, unchanged.
- VirtualGlove 0.4.1 is now the oldest supported in-place upgrade. Current player data, calibration, tuning, registries, pairing, and device settings are preserved. Older formats are left untouched and receive a clear unsupported version message.
- Rapid A/B now defaults on only where an individual Mattel program description explicitly identifies a rapid or pulsed button action: Program 7 A, Program B A, Program H A/B, and Bad Street Brawler B. Every other profile defaults off; documented compound, directional-pulse, and turbo actions remain independent.
- Program H now follows the documented finger order: thumb produces pulsed A and index produces pulsed B. Program B also pulses its documented horizontal movement as well as its flap action.
- Program 12 now holds A for the full thumb curl by default so Super Mario Bros. jump height remains controllable. If Rapid A is explicitly enabled, it repeats 250 ms A holds separated by the standard short rapid-fire gap.
- Clarified the distinction between the original glove's page-14 power-on state and VirtualGlove's source-accurate profile defaults. Documented that Rapid A/B changes only button repetition, while profile-owned fast turns, pulsed movement, turbo movement, and compound actions keep their own timing. Added upgrade guidance for preserved per-game overrides and **Use profile defaults**.

Removed

- Removed abandoned motion experiments, duplicate web code, obsolete camera commands, and settings that no longer affected gameplay.
- Removed the old unsigned Controller network format. VirtualGlove now accepts only the authenticated version used by current releases.
- Removed upgrade code for versions older than 0.4.1. Current player data and native Power Glove emulation names remain unchanged.

[0.4.1] - 2026-09-13

This release added the original numbered Power Glove programs, rebuilt the Gameplay Guide around them, and made hand centring and joystick dead-zone setup much easier to see and test. It also completed the visible VirtualGlove rename on the UNO Q and RetroPie while preserving existing player data.

At a glance:

- Programs 1-14 joined Programs A-I and the game-specific profiles.
- The camera test gained a live joystick grid, hand centring, and immediate dead-zone previews.
- The optional Ready-to-Play guide walks a player from connection to a running registered game without sending controls during practice.
- Fresh installations start with **Gestures off**.

- Service and application names changed to VirtualGlove while existing pairing, players, calibration, tuning, and games were preserved.

Added

- Added exact Program 1-10 filename aliases observed in the NES library for revision-tagged, alternate-region, and `Robo Warrior` spelling variants. These aliases retain the official profile and rapid-fire behaviour.
- Added the original Power Glove Programs 1-14 as first-class profiles, with their documented movement, finger, wrist, depth, compound-action, rapid-fire, centring, and manual-control behaviour. Added Mattel's indexed game mappings, exact case-insensitive `.nes`, `.zip`, and `.7z` aliases, and per-game rapid A/B exceptions without changing Programs A-I or the dedicated game profiles.
- Added validated structured game-registry entries with optional `rapid_a` and `rapid_b` switches. Older simple game entries still work, and the selected rapid-fire choices follow the active game session.
- Added Program 1-14 numbers to the Controller's Matrix display without changing the existing letters or game-specific symbols.
- Expanded the Game and Gesture Guide with gesture-by-gesture Programs 1-14 cards, compound-action timing and release rules, the complete official game index, rapid-fire exceptions, physical-controller use, and Program 14's camera-off behaviour. Synchronized the installation, configuration, matrix, architecture, native-emulation, quick-reference, and project guides. Updated Pixel Pal's Extra-Digit Hunt for the added illustrations; its verified answer is now 25.
- Added a consistent **Profile / See it** matrix preview to every Program 1-14 and A-I card. Numeric previews are generated from the exact one- and two-digit glyph placement used by the UNO Q matrix firmware.
- Reorganized the Gameplay Guide into explicit Program-card sections for 1-14 and A-I, and added Programs 2, 11, and 13 to the off-script starter cards. Renamed the Super Glove Ball research-core wording to describe the supported native system component and clarified that only unused, unconfirmed raw packet codes remain neutral; all confirmed native game actions are mapped.
- Rebuilt the numbered-program presentation to match the guide's established game-card design, added a complete Programs 1-14 quick selector, and replaced numeric matrix art in the off-script table with representative hand controls.
- Added dedicated closed-fist push, pull, left-punch, and right-punch artwork so compound numbered-program controls no longer reuse open-hand depth drawings. The deliberately incorrect six-digit illustrations and all 25 appearances in Pixel Pal's Extra-Digit Hunt remain unchanged.
- Added a visual-only Vulcan-salute easter egg to live camera experiences. Holding an extended hand with a deliberate middle/ring split briefly shows **Live long and prosper** with Pixel Pal, then fades without changing or suppressing controller input. It requires a release before rearming, has a 30-second cooldown, stores no hand data, and announces the phrase politely to screen readers.
- Added **Centre hand** directly to the Joystick dead-zone camera test. It is available only while the safe camera test is running. Game output stays paused, and the grid redraws around the newly saved hand centre and hand size.
- Added a 3×3 camera grid to the dead-zone test, using the saved calibrated hand centre and the same translated, full-size bounds as gameplay. A subtle live direction region highlight clears on tracking loss or suppressed movement; no labels or palm marker are drawn. Moved the camera toggle beside **Use standard size**.
- Added an off-by-default camera test to **Setup -> Joystick dead zone**, with its mirrored preview and live direction feedback. It safely releases the camera when stopped or when the page closes.
- Added an optional, resumable **Get ready to play** guide at `/ready`, linked from Setup and Dashboard. It rechecks live readiness and uses safe practice for essential gestures before an explicit transition to registered-game controls.

- Added separate versioned per-player guide progress, lossless version-5 store migration, and a safety pause that remains in place if the guide is interrupted.
- Added a read-only Connection Doctor to **Connect to RetroPie**, with progress, separate address/service/authentication and runtime checks, plain-language next actions, and a sanitized downloadable checklist report. Unsupported game-side readiness checks remain explicitly unverified.

Changed

- Renamed installed RetroPie and UNO Q services, executables, runtime paths, configuration directories, and managed installation records from `powerglove-*` to `virtualglove-*`. Upgrades preserve pairing, games, launch settings, camera recovery, and rollback backups. Names that genuinely refer to Mattel's Power Glove or native emulation remain unchanged.
- Changed the fresh-install startup profile to **Gestures off**. Existing saved startup profiles remain unchanged during upgrades.
- Moved **Joystick dead zone** directly below **Players** in Setup so player selection, centring, and movement-box adjustment stay together.
- The joystick camera test now previews unsaved slider changes immediately in its grid, highlight, and direction pills, using the saved hand centre and current live palm position. Camera toggles retain the draft; player changes discard it. Saving is still required for gameplay, and feedback waits for updated worker bounds before returning to saved D-pad output.
- Ordered Setup with separate **Connection and startup** and **Pair with RetroPie** sections. **Show statistics** remains at the very bottom, after Games, and shares its default-off switch with Dashboard. Recent events is retained because it reports recognised gestures such as Glove Zap.
- Joystick dead-zone size now means a chosen width and height fraction of the full camera frame, anchored to each player's saved neutral palm centre in both preview and gameplay. The effective square is at least 1.5 times the saved calibrated hand size; near a frame edge it translates inward without clipping or shrinking. Live hand size and jitter never make the box breathe. All existing numeric choices and player/calibration data remain unchanged, as do native Super Glove Ball X/Y calibration and reach.

Fixed

- Distinguished **Gestures off - no active profile** from **Program 14 - Physical controller only** throughout Dashboard and the guides. Program 14 now reports rapid A/B off and keeps its steady **14** matrix profile display while its camera and VirtualGlove output remain closed. Both states leave RetroPie's merged physical Player 1 joypad available.
- Made fresh installations emit only `virtualglove-*` service, executable, runtime, and installation names. Upgrades move existing pairing, games, players, calibration, tuning, Academy progress, and camera recovery forward before placing obsolete files in a recovery backup.
- Fixed RetroPie upgrades from pre-0.4.1 installations so the bootstrap recognises the existing legacy launcher before asking for a Controller address. Package-source preflight now completes before the installer creates or migrates VirtualGlove configuration, and existing cabinet hooks are validated before that migration begins.
- Stopped old timers and background helpers in the correct order during an upgrade, preventing warnings and overlapping old and new services.
- Delayed UNO Q service migration when a shutdown is already pending, preventing the board from stopping halfway through an update.
- Changed engineering Wi-Fi deployment to include and checksum-verify the compiled Matrix image, flash it before application restart, require a matched firmware handshake, and use administrator access when retiring a root-owned legacy application directory.
- Strengthened release checks so a package cannot mix old and new service or executable names.

The entries below are the development workshop log leading to earlier stable releases. They retain more measurements and implementation names because they show why an experiment was accepted or rejected. A player can safely skip to the next stable-version heading; a tinkerer can follow the trail.

[0.4.0-rc.7] - 2026-09-11

Added

- Added friendly first-install Controller naming with `virtualglove.local` as the suggested address, custom-name validation, confirmation, visible LAN conflict detection, an explicit unattended option, and upgrade-safe hostname preservation. The containerized HTTPS server now uses the approved UNO Q host identity rather than a transient App Lab container name.
- Added a persistent per-Controller local certificate authority, HTTPS-only public trust-certificate download, automatic named leaf renewal, and guided one-time browser trust while retaining physical Matrix fingerprint pairing.
- Added clear installation and troubleshooting guidance for RetroPie systems whose Raspbian Buster package source has moved to the legacy archive.
- Rebuilt the optional Engineering Toolkit as a curated, self-checking release archive with a dedicated technical guide, repeatable environment setup, categorized commands, clean-extraction validation, and resolved-package records.

Changed

- New Controller installations now use `/home/arduino/ArduinoApps/virtualglove`, so App Lab creates only `virtualglove-*` containers and networks. An existing pre-rebrand installation is stopped without relaunching, has its private data copied safely, and is moved into the installer recovery backup only after VirtualGlove starts successfully.
- Renamed the Controller's App Lab Compose project and running containers to `virtualglove-*`. Upgrades stop and remove the legacy project before starting the renamed containers, while protocol identifiers remain compatible.
- Extended RetroPie's single-use pairing-code window from two minutes to five minutes and clarified that the command must run on the exact console selected in Setup when multiple RetroPie systems are online.
- UNO Q release packages now carry checksum-described, precompiled Matrix firmware. Ordinary installations use the board's factory OpenOCD support and no longer download the Zephyr compiler toolchain. Source builds remain available in the full repository.
- RetroPie installation detects the obsolete Buster package host before making VirtualGlove changes and tells the operator exactly which source file needs repair without silently changing operating-system repositories.
- Removed release building, deployment, documentation generation, firmware stamping, and other incomplete repository-maintenance workflows from the Engineering Toolkit archive. Added missing direct helpers and kept normal installation and calibration documentation free of engineering setup steps.
- Made Toolkit archives reproducible, required side-effect-free help from shell tools, verified recorded environment identity, included the Toolkit PDF in Controller packages, and added an extracted-package CI workflow smoke test.
- Made zero-jitter motion simulations safe and changed the Toolkit's unsmoothed comparison to exercise the real latest-coordinate path.

Fixed

- Prevented release upgrades from starting App Lab's regenerated legacy Compose project while the renamed `virtualglove` project still owns its published ports. The installer stops both possible project identities before regeneration, then completes the existing one-stack migration.
- Added a transparent web-specific VirtualGlove logo so the Controller header blends with its page background while PDF covers retain the original artwork.
- Fixed clean UNO Q release installation after App Lab startup by explicitly preserving the application root while recreating generated Compose services; brick mounts can no longer collapse to invalid root-level `/scripts` paths.
- Prevented the release installer from generating a Python bytecode cache inside its verified staging tree before applying the installation manifest. RC.7's archive was clean, but this runtime cache caused a clean UNO Q installation to reject the installer's own generated path. Release packaging now exercises setup loading against each extracted package and rejects any staging mutation.
- Made every primary UNO Q download command enter `/home/arduino` before saving the installer. This prevents first-time installation from failing when the Controller terminal opens in the non-writable `/` directory.
- Corrected Pixel Pal's Extra-Digit Hunt answer from 14 to 15 after a complete visual audit found one more six-digit glove in the opening Rock, Paper, Scissors table. The documentation check now also rejects six-digit artwork that is labelled for accessibility but missing from the hunt manifest.
- Restored complete Python 3.7 release validation: camera-profile recovery and backup rotation now use compatible runtime APIs, and affected tests avoid newer mock-call conveniences. The release gate now runs the full suite on both Python 3.7 and Python 3.12 before packaging.
- Made local HTTPS certificate renewal independent of OpenSSL's version-specific hostname-check exit behaviour, so a changed Controller IP is always reflected in the trusted certificate. Corrected the remaining Python 3.7 Matrix test compatibility issue found by the release gate.

[0.4.0-rc.6] - 2026-09-11

Changed

- Reworked the repository README into a concise project landing page with a clear introduction, feature summary, hardware checklist, seven-step release- candidate installation path, first-play guidance, and simpler routes into the user and technical documentation. Removed the Extra-Digit Hunt explanation from the README while retaining the game in the illustrated guides.
- Added automatic paired-console recovery when a saved IP becomes stale or `.local` resolution is unavailable. The Controller now detects missing authenticated receiver replies, broadcasts only a signed handshake on each connected physical LAN, and resumes with the responder that proves possession of the existing pairing key. Controller states are never broadcast, and the recovery does not replace or expose the saved key.
- Added the reverse recovery path for RetroPie's registered-game profile renewals. A failed saved Controller destination triggers a ROM-free signed discovery request; only the paired Controller's request-matched response is accepted before the real profile command resumes by unicast. The authenticated address cache is bounded, lasts 30 seconds, and never rewrites saved configuration.
- Setup now distinguishes its saved console destination from the active authenticated controller link and can download a field-allowlisted system report. The report includes useful version, camera, controller, profile, and connection health while excluding frames, secrets, player/calibration data, ROM names, and network addresses.
- Added a fresh-hardware acceptance checklist covering clean installation, pairing, persistence, receiver restart, DHCP/name recovery, camera reconnect, coordinated upgrade, and final FCEUmm/native play confirmation.
- Consolidated controller-state cadence, signed handshake maintenance, paired-console discovery, registered-game leases, Setup health checks, host link sampling, recovery windows, and representative LAN traffic in one technical architecture table. The configuration and security references now link to that common operational explanation.
- Renamed the public GitHub repository to `VirtualGlove` and updated documentation and installer download sources to use the new canonical address. GitHub's old repository links remain usable as redirects.
- Clarified that guarded camera recovery and the `uhubctl` package are installed automatically by the standard Controller installer. The standalone recovery script is retained only for repair and development deployment.
- Simplified the first-time installation language, standardized Glove Academy naming, corrected post-rename checkout paths, and clarified that every native action required to complete Super Glove Ball has been confirmed in live play.

[0.4.0-rc.5] - 2026-09-11

Changed

- Renamed the project and all user-facing product identity to **VirtualGlove**, including the application, website, virtual controller, documentation, public PDFs, installer packages, screenshots, and release artifacts.
- Reworked the existing hand, camera, framing, and target logo into a single **VIRTUALGLOVE** wordmark. The Dashboard intentionally retains the quotation "I love the Power Glove. It's so bad." while its product description now says "your camera-only VirtualGlove."
- Retained the compatibility identifiers then needed by installed systems and native emulation, including legacy installation paths, service/script filenames, protocol version, and `lr-nestopia-powerglove` core name. These identifiers upgraded in place and were not presented as the product name.
- Renamed new portable hand-setup exports to `<player>-virtualglove-hand-setup.json` with format `virtualglove-hand-setup` version 4. Existing `powerglove-hand-setup` version-2 and version-3 files remain importable.

[0.4.0-rc.4] - 2026-09-10

Added

- Added five context-specific Pixel Pal stances for coaching, play, inspection, safety, and success. The website and friendly guides now choose his pose by purpose while preserving the existing welcome and Glove Master trophy art.
- Added a per-player square centre-box control for FCEUmm joystick movement, including all four diagonals, immediate centre release, live direction feedback, and automatic neutral-jitter protection.
- Added an Engineering Journey that records the one-week development process, measurement discipline, CPU/GPU investigations, discarded movement paths, reliability work, and lessons behind the current architecture.
- Added the complete GNU GPL version 2 licence text required by the distributed Nestopia-derived patch and clarified exactly which third-party components are bundled, built on the target, installed from Debian, or supplied by RetroPie.
- Added an on-demand, mirrored live view with centre and edge guides to Pixel Pal's camera-settings test so players can verify hand placement during sweeps.
- Added Pixel Pal's **Find the best camera settings** wizard to Setup. It safely compares the current camera configuration with supported reader, frame-rate, buffer, and exposure choices; rejects reader fallbacks and unsupported rates; protects hand continuity before ranking latency; and saves only an accepted recommendation for the same physical camera. Trials retain aggregate timing only-never video or images-and restore the exact original configuration after completion, cancellation, or an interrupted Controller restart.
- Added a persistent **Stop test** recovery control to the camera wizard. It remains available after a camera disconnect, restores any pending exact pre-test configuration, restarts normal vision, and returns the wizard to a clean state so the camera test can be started again.
- Added an explicit one- or two-buffer camera setting to Setup. Manual camera saves and Pixel Pal's camera recommendation now expose the same validated capture-buffer choice.

Changed

- Completed the MediaPipe CPU-pipeline refinement and froze the validated production defaults: OpenCV with thread isolation, one requested buffer, 30-fps-first negotiation, four inference threads, tracking confidence `0.35`, palm-detection confidence `0.45`, a 2.25 search region, direction-aware fast-sweep tracking, and Latest-coordinate native X/Y.
- Added detector-context attribution to the output-paused camera benchmark. A final sustained run separated camera dequeue, MJPEG decode, inference pickup, the preceding frame, palm detection, recovery, post-graph work, and Linux scheduling. Ordinary landmark tracking measured about 37 ms median; palm detection/reacquisition measured about 105 ms median. Camera delivery, scheduling, recognition output, and transport were not the source of the detector tail.
- Rejected lower `0.10` and `0.20` tracking-confidence gates. They did not prevent any fast-sweep detector entries and increased reacquisition p95 from about 102 ms at `0.35` to 126-128 ms. A broader clip likewise found no useful recovery trade-off.
- Extended process-isolated capture research to OpenCV as well as Direct V4L2. Both variants retain one coherent latest frame and report an explicit fallback; matched live tests still selected thread-isolated OpenCV for production.
- Made Controller restarts terminate and reap the complete worker process group, including its private `uv`, Python, and camera-owning descendants, before a replacement worker starts.
- Replaced positional activation/release hysteresis with the original-style nine-region layout: centre, four cardinal directions, and four diagonals. Every fresh position is classified independently; native Super Glove Ball X/Y, Menu Guard, tracking-loss safety, and Program-specific mappings are unchanged.
- Upgraded the player store to version 5 and portable hand-setup backups to version 3. Older stores and version-2 backups migrate the largest directional activation value into the new scalar centre box and discard obsolete releases. Positional directions are no longer offered as gesture-personalisation channels.
- Reworked Architecture as a current-system reference and moved historical movement, runtime, GPU, and reacquisition discussion into Engineering Journey.
- Consolidated the Programs A-I handbook into the Gameplay Guide, reordered its discovery flow, and retired the redundant standalone Programs PDF.
- Updated Installation, Camera, Configuration, Native Emulation, and Super Glove Ball Native guides for the current OpenCV default, optional Direct V4L2 path, one/two-buffer choice, Pixel Pal camera test, 180 ms X/Y-only brief-loss hold, and MediaPipe 0.10.35 CPU runtime. First-time setup now emphasizes safe defaults and links to detail instead of presenting driver choices as required.
- Extended only native continuous X/Y's brief missed-observation hold from 120 to 180 ms, covering roughly one additional MediaPipe result during an extreme corner-to-corner sweep. Buttons, finger states, depth, roll, and D-pad output still release on the first missed native observation. MediaPipe tracking confidence, direction-aware search, and the validated reacquisition logic are unchanged.
- Extended the output-paused camera benchmark for sustained UNO Q runs. It now separates driver dequeue age, MJPEG decode, recognition pickup, graph time, and post-graph work; learns camera sequence-counter cadence; records compact correlated tail events; and can attribute palm-versus-landmark paths without retaining images.
- Added a three-thread research lane and rejected it after a live four/three/four comparison. Three threads produced fewer coordinate results and higher normal latency without improving end-to-end p95, so the validated four-thread production setting remains unchanged.

- Added a benchmark-only process-isolated camera lane with Linux task-scheduler counters. A 90-second A/B/A comparison and five-minute soak showed that it continued draining frames through long MediaPipe palm-search calls, reduced the worst observed coordinate age from 303-327 ms to 239 ms, and completed without a camera error. It is not yet a production capture mode.
- Added process-isolated Direct V4L2 as an engineering comparison. It owns one replaceable shared frame, reports child failure repeatedly so existing recovery can act, restores manual exposure in the camera-owning process, and falls back to threaded OpenCV when startup fails. Matched live Super Glove Ball tests selected OpenCV with thread isolation as the production default: it felt faster, remained playable through fast sweeps, and recovered quickly after intentional departures from the camera view.
- Matched the isolated reader to the proven camera lifecycle by treating a camera-marked invalid MJPEG frame as transient. Hardware validation confirmed process startup, manual exposure 78/gain 96, 569 advancing driver frames over ten seconds, graceful shutdown, descriptor release, and automatic-exposure restoration. A deliberately killed child can still wedge this Kiyo/hub at the USB level, so destructive crash injection is not part of routine validation.
- Promoted the validated direction-aware fast-sweep search to standard MediaPipe behaviour. Removed its Setup checkbox and guarded save endpoint; older `directional_search` device-file values are now harmlessly ignored.
- Renamed the packaged CPU runtime from the misleading `powerglove.gpu2` local version to `powerglove.cpu1`. Its selected graph and behaviour remain the same; the new name makes clear that production inference uses XNNPACK on the CPU.
- Corrected Setup's Controller output flag so an armed Controller waiting for a game is shown as ready instead of falsely reporting **Receiver unavailable**. Genuine delivery failure remains red during an active game context.
- Made camera recovery refuse a whole-hub driver rebind when the enrolled hub also carries a network interface. Capability-confirmed per-port power cycling remains allowed; unsupported network-bearing hubs now fail safely instead of disconnecting the Controller's Ethernet path.
- Made worker startup probe its retained `uv` environment offline first. A complete cache now restarts without Internet access, while a new or incomplete installation automatically retains online dependency resolution.

[0.4.0-rc.3] - 2026-09-09

Added

- Added a separate, version-matched VirtualGlove Engineering Tools source archive for protocol traces, replay analysis, benchmarks, GPU experiments, soak tests, and maintainer build tools. It contains no ROMs, recordings, credentials, device data, cached models, or compiled cores.
- Added an optional, ROM-free **VirtualGlove Calibration Test** to RetroPie's Ports list. Its separately built `lr-powerglove-dot` core displays the same guarded native X/Y used by Super Glove Ball, automatically holds and releases a native test profile, and gives players a simple centre, reach, edge, tracking-loss, and recovery check without changing any game's emulator.
- Added a camera-free post-inference benchmark with repeated statistics-off, statistics-on, and statistics-off lanes. It measures established-session UDP transmission, Dashboard housekeeping, full-iteration tails, and newest-only publication under a deliberately slow status consumer.
- Added capture-timestamp tracking-loss and reacquisition telemetry. Diagnostics now separate the duration since the last detected hand, the observed missing span, the recovery inference time, and the longest loss instead of inferring recovery cost from worker-loop timing.
- Added a production-matched, user-paced fast-sweep recorder and a replay-only zero/one-frame directional-search recovery comparison. Direct V4L2 captures retain their actual buffer, exposure, and gain settings, restore camera automation on exit, and retry brief invalid-frame gaps without abandoning the recording.

Changed

- Replaced the former MediaPipe 0.10.18 package with MediaPipe Hands 0.10.35 as the sole CPU recognition runtime after matched replay and live Super Glove Ball validation. Gesture rules, calibration, Latest-coordinate movement, and controller mappings are unchanged. The rebuilt ARM64 wheel removes unused JAX/JAXLIB dependency declarations and pins headless OpenCV 4.11.0.86.
- Focused ordinary Controller and RetroPie release packages on production code and end-user calibration, status, recovery, and emulator-setup tools. The full engineering suite remains in Git and on development deployments but is no longer installed for first-time users.
- Made the active emulator part of RetroPie's authenticated, renewable game heartbeat. Only the exact `super_glove_ball` plus `lr-nestopia-powerglove` pairing selects native packets; FCEUmm and every other or unknown core use joystick output. Changing cores is treated as a controller transition so held native or joystick state cannot leak between modes.
- Coordinated guarded camera recovery with the vision worker: the worker closes the camera before the enrolled hub reset, waits for the helper result, then reopens and retries only after its cooldown. Recovery applies to any active gesture profile, whether output is native or joystick. A headless Kiyo Pro test confirmed the guarded stop/reset/restart sequence and USB re-enumeration, but the camera still required a physical reconnect before it produced frames; USB presence alone is not reported as stream recovery.
- Added capability-detected camera-port power cycling through `uhubctl`. The helper targets only the enrolled camera's exact port when the hub is listed as per-port switchable and never forces unsupported hardware. Older enrollment files and unsupported hubs retain the identity-checked whole-hub rebind. Helper completion now means only that the USB action ended; recovery is confirmed separately after the restarted worker receives a real frame.
- Kept Dashboard work out of the controller-critical boundary. Tuning state is resolved before MediaPipe inference, gameplay state is sent before periodic handshake maintenance, browser status preparation is limited to 10 Hz while control transitions publish immediately, and preview mirroring no longer selects a slower MediaPipe preparation path.
- Bounded handshake reply processing to two datagrams per frame and replaced a recursive controller-state copy with an explicit signed-wire mapping. These changes preserve authenticated delivery and receiver-restart recovery.
- Aligned the example Controller configuration with the shipped four-thread, 0.35 tracking-confidence and 2.25 search-area defaults.
- Made Latest coordinate the only live Super Glove Ball native X/Y behaviour. Removed the Dashboard movement selector and its restart endpoint; older saved bounded-mode values are ignored, while historical benchmark tooling remains available for engineering comparison.
- Reorganized Setup into shorter, action-first sections. Tracking and controller output now use the same red, grey, and green status flags as connection checks; the console-check age and gameplay caveat remain visible without the redundant colour legend.
- Put player naming controls before their explanation, summarized what each player and hand-setup backup contains, and named downloaded backups after the player, such as `alex-powerglove-hand-setup.json`.
- Separated Camera from Connection and startup, moved pairing-key and live camera status text beside their related controls, and moved joystick dead-zone help below its controls.
- Added a focused Camera guide covering automatic selection, frame rate, capture reader, exposure, lighting, reconnection, and troubleshooting.

Validation

- Replayed the same 736-frame fast-sweep clip through both packaged runtimes. Detection continuity and all 15 one-frame misses were identical at 97.96%, while MediaPipe 0.10.35 reduced overall inference p95 from 60.18 ms to 46.60 ms and palm-reacquisition p95 from 149.48 ms to 120.16 ms. A live Super Glove Ball trace delivered all 3,408 samples, improved capture-to-send p95 from 126.84 ms to 91.16 ms, and had zero trace drops. A ten-minute output-paused soak completed without errors, reached 61.7 °C at its hottest reported zone, and settled at a stable memory plateau.
- Corrected the saved-clip loss interpretation after frame-level review. Its nine-frame run is the scripted hand-removal cue, while the selected fast-sweep lane detected 91 of 92 frames and missed only the exact cue-boundary frame. Replay reports now label missing runs with frame ranges and cue names.
- Replayed the saved 402-frame sweep with Dashboard preview closed and open. Both lanes used identical fused preprocessing and 96.52% detection continuity; inference p95 was 4.70 ms closed and 4.65 ms open on the development Mac.
- A 100,000-iteration synthetic soak kept signed UDP send p95 at 0.0179 ms in all three off/on/off lanes. A deliberately 5 ms status consumer did not block controller sends because pending Dashboard snapshots were replaced newest-only.
- Replayed a current-settings Kiyo Pro fast-sweep clip through the production four-thread lane. The selected `2.25` search area, `0.35` tracking confidence, and `0.55` palm-detection confidence retained 98.82% fast-sweep detection. Larger search areas, lower detection thresholds, and a one-frame search-offset carry did not improve the moving cue, so immediate reset and current defaults remain unchanged.
- A matched exposure-78/gain-96 lighting comparison improved overall detection continuity from 98.73% to 99.57%, reduced missing frames from three to one, and reduced reacquisitions from two to one. Fast-sweep detection remained effectively tied at 98.82% and 98.84%, while inference stayed near 35 ms median and 49 ms p95. Better front lighting adds tracking margin but does not make MediaPipe inference itself faster.

[0.4.0-rc.1] - 2026-09-08

The first public 0.4.0 release candidate combines the validated Latest-coordinate native movement path with lower-overhead capture, clearer camera selection, portable exposure defaults, and guarded recovery. It is suitable for interested users to install and test, while remaining a prerelease rather than the final 0.4.0 release.

Changed

- Replaced Setup's free-form camera number with a live dropdown containing Automatic and the currently discovered usable cameras. The list refreshes while Setup is open, excludes codec-only video devices, and keeps a saved but temporarily disconnected selection visible instead of silently changing it.
- Added an optional capability-checked Manual exposure and gain setting for the Direct V4L2 camera path. Automatic remains the portable default. The worker applies controls through the active stream, reports requested and actual values and supported ranges, falls back explicitly when unsupported, and restores automatic exposure when vision closes. A matched Kiyo Pro play test selected exposure `78` and gain `96` for this Controller because continuity was slightly better than Automatic while measured latency was unchanged.
- Hardened camera recovery by distinguishing a healthy-camera enrollment from a sustained stream-recovery request. The allowlisted hub can now receive one guarded reset when video is wedged even if the camera still appears in USB enumeration; ordinary reconnects continue to update the saved camera-to-hub association.
- Promoted MediaPipe's `2.25` next-frame hand search area and `0.35` tracking confidence as the reproducible defaults. The larger search area recovered five of nine missed fast-sweep frames without material latency or false-activation cost. Palm-confidence changes, every-frame palm detection, and the full landmark model did not pass their latency/continuity gates and remain benchmark evidence rather than gameplay modes.
- Added advance preparation for the next physical hand-to-display latency session: a privacy-safe read-only two-device preflight, a short framing/smoke protocol, a reversible Controller/receiver trace manager, and contact-sheet-based video review. These tools leave movement math unchanged and cannot treat terminal cue time or unrelated device clocks as physical latency evidence.
- Moved Dashboard status publication onto a latest-only worker. Changed controller state is submitted immediately; routine detailed gesture and controller feedback is limited to about 10 Hz, and rolling percentile summaries to 2 Hz, only while **Show statistics** is enabled. Ordinary camera preview no longer requests landmark diagnostic payloads merely to draw the already-available skeleton.
- Added an opt-in direct V4L2 capture backend for Linux 64-bit, 640×480 MJPEG cameras. It drains to the newest driver buffer, retains a valid monotonic camera timestamp, and automatically reopens the compatible OpenCV path if the camera or negotiated format is unsupported.
- Added capability-detected low-latency exposure choices. Standard UVC controls are queried before any write; supported cameras retain automatic exposure but can disable exposure-driven frame-rate variation. The Kiyo Pro choice also requests the existing USB-identity-checked volatile HDR-off mode.
- Added a configuration-only lean MediaPipe Hands comparison that requests image landmarks without world-landmark or handedness output streams. The full graph remains the production default until matched Controller tests prove the lean output set is faster without recognition regressions.
- Consolidated overlapping documentation by purpose. Third-party licensing, runtime provenance, asset origins, and modified-core distribution now share one notice; screenshot and web-art production moved into Contributing; the early-start repair material moved into the Configuration Reference; and the September 6 Setup review was folded into this release history.
- The September 6 review covered Setup routes, persistence, worker controls, pairing, signed delivery, installation manifests, player backups, hostname refresh, networking indication, and web-module separation. Automated checks, browser interaction tests, package verification, and live paired-device checks passed for the recorded fixes. Its then-outstanding camera-to-display latency work is now tracked in the movement evidence documents rather than a separate review file.
- Regenerated the complete 19-guide PDF set and Controller Help from the consolidated Markdown structure.

- Added end-to-end architecture and timing-boundary diagrams showing the shared camera and MediaPipe front end, authenticated Controller-to-RetroPie delivery, the FCEUmm and custom Nestopia paths, and the final game/display response. Camera benchmark results now state exactly which part of that chain each test measured and what remained outside it.

Validation

- The final release-candidate suite passes 535 tests with one expected Linux-specific `IP_PKTINFO` integration skip on macOS. Documentation audits cover all 19 Markdown guides and matching PDFs, and browser interaction tests cover live camera discovery, camera selection, exposure choices, and saved Setup behaviour.
- A live Razer Kiyo Pro test accepted the direct V4L2 path at 640×480 MJPEG and 30 fps. The driver supplied monotonic timestamps and advancing sequence numbers; a representative fresh sample measured about 1.25 ms from the driver timestamp to userspace dequeue, with the expected 33.3 ms capture cadence. Direct V4L2 remains optional because other Linux cameras and drivers may negotiate differently.
- The output-paused lean-graph comparison was not promoted. Against the same direct-capture setup, its median inference time improved only from about 43.9 ms to 43.2 ms, while p95 sample age increased from about 108.5 ms to 114.6 ms. The complete graph therefore remains the default.

0.4.0 development baseline - September 7, 2026

This unpublished development baseline is included in `v0.4.0-rc.1`. It promotes the native-movement efficiency work validated after the 0.3.2 release candidates. The shared gesture mappings and FCEUmm behaviour remain compatible; the principal change is a lower-latency, more precise MediaPipe path for native Super Glove Ball X/Y.

Added

- Added Automatic, 30 fps, and 60 fps camera-rate choices under Setup's advanced camera settings. Automatic tries the tested 30-fps path first, falls back to a driver-selected usable rate, and reports the live negotiated rate.
- Added an optional **Show statistics** preference to Dashboard and Setup. It is off by default, stored only in the browser, and stops the page from reading, rendering, or retaining detailed controller, axes, finger, performance, and recent-event fields while disabled.
- Added native trace evidence for raw MediaPipe observations, detection gaps, recovery decisions, and one-result confirmations. The analyzer can now review ordinary Latest-coordinate traces without enabling optical-flow diagnostics.

Changed

- Made **Latest coordinate** the production native X/Y default. Continuous tracking publishes each newest valid, reach-clamped MediaPipe coordinate directly. **Bounded speed curve** remains available as an explicit comparison.
- Changed the Controller baseline to four MediaPipe inference threads, a `0.40` tracking-confidence threshold, and 30-fps-first camera negotiation after matched live tests. New installations receive these values; saved device settings remain preserved during updates.
- Moved preview landmark drawing, status text, downscaling, and JPEG encoding off the inference thread. Gameplay previews use a 320×240 latest-only copy while Academy and tuning retain full-size visual feedback. Superseded preview work is discarded instead of delaying controller delivery.
- Avoided calculating unused experimental palm anchors during ordinary gameplay. Diagnostic runs still expose the complete candidate set for comparison.
- Preserved a player's valid comfortable-reach spans when **Centre hand** updates neutral pose and jitter. If a moved centre makes the saved endpoints unsafe, the mapping falls back to the full camera field instead of saving invalid reach.

Fixed

- Prevented the backward-then-forward Robo-Glove jump seen after brief MediaPipe reacquisition. Latest-coordinate mode now accepts strongly aligned forward recovery immediately, but holds one contradictory or unusually distant non-forward result until the next fresh measurement. It does not predict, smooth, queue, or overshoot continuous movement.
- Ensured the newest captured frame and its capture timestamp remain authoritative through inference, freshness checks, native mapping, tracing, and transmission.
- Corrected the staggered-tracker benchmark to compare matched one-by-four and two-by-two CPU lanes using real sidecar capture timestamps. The staggered lane was rejected because it increased source age, reduced continuity, and produced a much larger coordinate jump.
- Preserved advanced measured device settings when Setup saves ordinary connection or camera fields instead of reconstructing a smaller configuration document.

Validation

- The complete Python suite passes 496 tests with one expected Linux-specific skip. Browser interaction coverage includes camera-rate persistence and the statistics preference's default-off, safe-rendering, cross-tab, and no-work behaviour.
- Live Super Glove Ball tracing measured about 65 ms median camera-to-coordinate age. In the confirmation window, five strongly forward recoveries passed immediately and the new guard inserted no unnecessary holds. The player reported the tightest, most hand-attached response reached during testing.
- The rejected staggered-tracker comparison measured 20.23 Hz and 45.64 ms p95 source age for the one-by-four baseline versus 21.13 Hz and 66.03 ms for the two-by-two lane; detection continuity fell from 99.14% to 98.09% and maximum coordinate step rose from 0.0407 to 0.3447.
- An isolated Adreno GPU delegate experiment remains non-production: the tested MediaPipe Tasks graph was substantially slower than the proven CPU MediaPipe Hands path. Version 0.4.0 ships no custom GPU runtime.

[0.3.2-rc.8] - 2026-09-07

Changes since rc.7: per-player movement reach, selectable latest/bounded native X/Y, corrected capture timing and landmark validity, edge-safe recovery, lower native publication latency, and isolated acceleration research tools. The project base version remains 0.3.2. This is a prerelease.

Added

- Added a separate Glove Academy **Movement reach** editor for the active player. It exposes left, right, up, and down normalized spans, reports tracking-area dimensions and aspect ratio, saves only those four fields, and can restore the full-camera mapping without changing centre, gestures, or lesson progress.
- Added Dashboard selection between **Bounded speed curve** and **Latest coordinate** for native Super Glove Ball X/Y. Both modes use the same MediaPipe observations, calibration, reach mapping, and safety behaviour.
- Added a four-lane native movement comparison for version-2 vision replay reports. It compares the former overshooting experiment, a safely capped error-driven reference, the bounded speed curve, and direct latest coordinates; a deterministic 27-candidate sweep reports jitter, lag, medium response, fast pickup, reversals, overshoot, continuity, and source age when available.
- Added output-paused benchmarks for palm-anchor stability, mirrored-frame preprocessing, one/two/four inference threads, tracking confidence, two staggered trackers, and MediaPipe Tasks live-stream CPU/GPU experiments. These tools do not create a production GPU mode or change gameplay output.
- Prepared optional per-frame motion traces separating recognised, flow, selected and filtered coordinates, including source freshness and fallback reasons. Added an offline saved-sample review and an actual-engine smoothing step model; the model is not a physical latency measurement.
- Added `analyze-motion-trace.py` and `compare-motion-matrix.py` for normalized movement classes, selected-versus-filtered error, settling estimates, source-age distributions, fallback reasons, and tracking-loss counts. Ran the six trace-only smoothing configurations on the UNO with zero dropped trace records and restored the live setting afterward.
- Added Setup -> Joystick dead zone: a per-player size slider with automatic half-distance release, live direction indicators, and a standard-size preset. Saves update all four digital direction thresholds while preserving centre, native reach and other gestures. Advanced directional pairs remain in Glove Academy.
- Added an optional UNO Q test workflow for the cabinet's installed dot core, dot-labelled guided status sessions, and a read-only native-state probe for validity, loss/recovery and coordinate ranges. MediaPipe, controller output and game defaults remain unchanged; physical comparison is pending.
- Benchmarked Kiyo Pro capture on UNO Q and added an opt-in 640×480 MJPEG/two-buffer/volatile-HDR-off candidate, which delivered 59.7-59.8 fps in isolated capture repeats. Higher 720p decoding costs ruled out copying the Pi resolution. Inference threads are unchanged; recognition-under-load and physical latency validation remain pending.
- Ported optional per-player comfortable reach spans from the Raspberry Pi version. Both MediaPipe response modes map asymmetric reach to the screen edges, player backups preserve spans, and re-centring clears them. Added a guided, output-paused calibration helper using raw palm measurements in practice mode. Camera defaults and inference threads are unchanged pending UNO Q measurements.
- Added and evaluated an experimental Super Glove Ball optical-flow path. Live testing found it jerky and unreliable, so it is now archived as research code and historical trace support rather than offered as a runtime option.

Fixed

- Restored Python 3.7 compatibility in historical trace and browser-support tools and the optional capture trace's thread identifier. The capture-thread test also allows heavily loaded CI runners more scheduling time. These changes do not affect production capture timing or controller output.
- Native speed and freshness calculations now use the selected camera frame's capture timestamp instead of inference-start time. Invalid or non-finite landmark geometry is rejected before mapping, while MediaPipe's handedness score is no longer misrepresented as a position-confidence score.
- Both native X/Y modes clamp their input to the active player's calibrated reach before response processing. Leaving an edge pins the Robo-Glove there without building hidden off-screen filter state; tracking recovery starts from the first fresh clamped coordinate.
- Reworked bounded X/Y as one coherent vector response with an elliptical calibrated-noise region. Small axis noise no longer causes independent snaps; meaningful reversals are immediate, and stops settle inside the noise region on the first fresh result and exactly by the next.
- For native Super Glove Ball packets, the RetroPie receiver now publishes the native state before updating the unrelated virtual gamepad. Correlated traces report socket-return-to-native-publication and native-write time separately.
- Made MediaPipe Hands the sole live coordinate authority for native Super Glove Ball X/Y. The bounded curve and direct-latest mode affect only coordinate response; FCEUmm and gesture recognition are unchanged.
- Prevented brief MediaPipe dropouts from making the native glove jump through neutral and back. The last X/Y can be held for up to 120 ms while action, finger, depth, roll, and D-pad state releases immediately; sustained loss still neutralizes the complete native sample.
- Corrected the experimental optical-flow Dashboard marker to use the full preview dimensions instead of the downscaled 320-pixel work image, eliminating its upper-left visual offset.
- Replaced native Super Glove Ball's error-driven catch-up and optional extrapolation with a calibrated-reach speed curve. Resting jitter uses a per-player noise floor, deliberate movement becomes progressively more direct, and stops or reversals accept the newest measurement immediately. Compatibility settings above `1.00` can no longer overshoot.
- Pairing now reports success only after RetroPie answers a signed controller handshake with the newly installed token. This catches a copied-but-unusable token while keeping controller arming and actual emulator input as separate checks.
- Experimental X/Y now falls back directly to fresh, confident MediaPipe coordinates when source-to-current flow correction fails or exceeds its budget. Failed flow is cleared, fallback reasons are reported separately, and the original 250 ms recognition freshness limit remains enforced.
- Bound experimental X/Y correction to one checked source-to-current flow step on smaller images, and dispatch the next recognition job before correction. Added correction/pickup/failure diagnostics and a synthetic before/after benchmark. The original MediaPipe path remains the default; live recognition and physical latency validation are pending.
- Restrict UNO host mDNS to detected physical network interfaces during host setup. The test UNO advertised Docker bridges and renamed itself to `ArduIain-2.local` after a conflict during container restarts, breaking profile heartbeat delivery. The backed-up interface correction restored the original hostname and automatic game-profile recovery across a verified app restart.
- Dashboard Centre hand and Start controller clicks now survive status refreshes in Safari/WebKit. Controller requests stay disabled while pending, and centring guidance names the selected player beside the controls. Request feedback is announced and displayed beside those controls.

Changed

- Kept the calibration-compatible five-point palm average as the production anchor while exposing four candidate anchors to replay analysis. A different anchor must prove at least 25% less pose-induced movement, retain at least 99% of deliberate travel, and preserve continuity before it can replace the existing coordinate contract.
- Cached expensive percentile telemetry at 2 Hz while preserving per-inference recognition state and controller publication. The Controller supervisor now passes its validated one, two, or four-thread choice explicitly; the tested default remains two.
- Corrected the bounded native curve for the proven MediaPipe backend's measured 9-10 Hz cadence. The previous 60 Hz follow reference converted the configured `0.70` slow-follow weight to nearly `1.00` at runtime, making ordinary native X/Y visibly step between raw landmarks. Saturated `1.0` jitter measurements in otherwise reusable calibration files now fall back to the small fixed noise floor instead of producing a large dead zone and jump.
- Added explicit `native_xy_source` diagnostics for `mediapipe` and `inactive`, plus `native_xy_mode` diagnostics for `bounded` and `latest`. Both modes retain the normal MediaPipe landmark preview.
- Synchronized the architecture, installation, security, troubleshooting, command reference, built-in Help, README and PDF editions with the asynchronous movement path, comfortable reach, Kiyo capture candidate, per-player joystick dead zone, motion-analysis tools and post-pairing token verification. Standardized new user-facing diagnostic titles on **VirtualGlove Controller** while retaining literal UNO Q filenames and hardware references.
- Added an optional experimental-only X/Y smoothing boost override. The UNO medium-jump trial uses 8 instead of 4, lowering the per-axis immediate-response threshold from roughly .075 to .0375 camera units without changing synchronous tracking or reach calibration.
- Previously added an experimental-only `motion_coordinate_max` cap for controlled extrapolation tests. The exploratory UNO setting used cap 1.30 with boost 15 and minimum smoothing 0.70. That experiment established the stop/reversal risk and is superseded by the bounded speed curve above; retained configuration fields no longer permit overshoot.
- Moved player creation, renaming, deletion, and hand-setup backup/restore into Setup -> Players. Academy and Dashboard offer compact selectors for the same Controller-wide active player. Dashboard places Player before Active profile and combines game name and session status in one Game card.
- Glove Academy player selection now immediately loads sensitivity, lesson progress, and the player's saved centre. Removed the separate Use player and Reuse my saved centre buttons. Players without a saved centre still need Centre hand; switching keeps controller output paused, and backup-import calibration reuse remains an explicit choice.
- Renamed the explicit centring action to **Centre hand** throughout Dashboard, Glove Academy, player settings, personalisation, and maintained instructions. Centring continues to save only to the selected player; switching players during a sample prevents it being saved to the new player.

Validation

- A live Super Glove Ball status run with the Dashboard closed measured 96.5% detected samples, 65.4 ms median and 76.2 ms p95 sample age, 52.2 ms median and 58.3 ms p95 MediaPipe inference, 2 ms p95 send work, and about 16 distinct native updates per second. The player reported that control was improving; this is not yet the synchronized physical hand-to-display acceptance test.
- The UNO Q's Adreno 702 was reached through EGL/OpenGL ES and a custom isolated MediaPipe Tasks build created a TensorFlow Lite GPU delegate. That first heavy Tasks graph measured about 664 ms warm p50 versus about 207 ms for the same Tasks path on CPU, far slower than the proven MediaPipe Hands path. The wheel and temporary runtime changes are not in this release; the Controller was restored to its clean CPU image. GPU capability is confirmed, while a lean GPU palm/landmark path remains untested.
- Confirmed the tested cabinet accepted the newly paired token and subsequently delivered Controller input to a running game. The pairing-complete signal remains deliberately scoped to receiver authentication; the follow-on game test establishes the rest of this installation's path.

The remaining release gate is a synchronized high-frame-rate recording of the physical hand and display. Software-stage measurements cannot establish camera exposure, emulator/display delay, or the complete hand-to-screen result.

[0.3.2-rc.7] - 2026-09-06

Changes since rc.6: guided Setup pairing, networking indicators, refreshed guides and screenshots, and opt-in latency measurement tools. The project base version remains 0.3.2.

Changed

- Gave the Security network-exposure table dedicated column widths so port numbers stay intact, with less space assigned to Boundary. Refreshed every documented application screenshot from current templates using isolated sample data and an omitted-camera placeholder; added a repeatable capture script and rebuilt the affected PDF editions.
- Pairing now releases the physical approval-PIN display after the submitted request succeeds or fails, allowing the normal matrix display to resume without waiting for the authorization timer. Idle animation still respects On, Dim, or Off. Setup now serves directly at `/setup`; removed the page-revision redirect and link rewriting while retaining the shared favicon and compatibility with old bookmarks. Refreshed Setup screenshots and maintained guides.
- Fixed a Safari/WebKit pairing click failure caused by replacing button text during the half-second refresh. Pairing now brings a dedicated progress panel, completion, and errors into view; missing credentials receive explicit guidance, and live announcements are no longer inside a busy region. Verified delayed mouse clicks and progress feedback in WebKit and Chrome.
- Replaced Setup pairing with three inline steps using the saved console: method selection, physical Controller confirmation, and RetroPie credentials. Retained one-time-code and SSH-password methods, added expiry/retry guidance and secret clearing, renamed the connection button to Save settings, and removed duplicate Controller/power actions from Setup. Pairing APIs and server security checks are unchanged.
- Ordered Help user manuals as This console, Game and gesture guide, Programs A-I, Matrix display guide, Build your own, Installation and setup, and Troubleshooting by symptom. Kept the native-emulation introduction beside the Super Glove Ball compatibility record in Technical documentation. Unified browser-tab icon selection around one versioned multi-size ICO shared by Setup and every other page. The temporary Setup page-revision workaround has since been removed; the plain Setup URL is used throughout.
- Shifted the idle glove's bottom two wrist rows one pixel right, including the cuff entrance and wrist spark. Refreshed the simulated animation and manual illustrations; this change requires a matrix firmware update. Added versioned website icon URLs to refresh cached icons, including Setup.
- Renamed the fourth connection marker to Networking and extended its host check to physical Wi-Fi or Ethernet, including USB dock Ethernet. Virtual interfaces are excluded; legacy telemetry remains readable. This follow-up requires a Controller app and sampler update, not a new four-pixel firmware or RetroPie deployment.
- Reorganized Gameplay, Configuration, and component documentation; made Academy learning and per-player backup file locations explicit across guides and READMEs. Clarified both pairing PIN flows, native packet bytes 7-8, and Menu Guard's native-position limitation. Standardized Gun Smoke display text while retaining exact filenames, added wrapping for audit/benchmark tables, and refreshed Help assets and PDFs. The documentation-review batch was deployed as an uncommitted development build; the subsequent Networking and community-guide additions were also deployed without a commit or release.
- Simplified the tracker display name to **MediaPipe Hands** in the interface, diagnostics, command help, and documentation. The `legacy` identifier and tracking behaviour are unchanged.

Added

- Community guides for building the hardware, understanding native Power Glove emulation, and troubleshooting by symptom, available in Help and PDF editions.
- Hand-and-target website icons for browser tabs and saved home-screen shortcuts, included in Controller installations and upgrades.
- Setup now starts with four labelled status markers matching the Off-mode pixels, plus tracking, controller output, and saved console details. Shared background checks distinguish failed checks from unavailable results and keep physical network-link health independent of RetroPie connectivity. Pairing wording makes the Controller matrix PIN requirement explicit for both methods.
- Native Super Glove Ball test tools for three stationary holds and ten movements per direction, with sampled tracking losses, gaps, and neutral button counts.
- Opt-in, bounded Controller/receiver timing traces and a separately built diagnostic native core. Normal transport, native-state format, and responsiveness settings are unchanged. No diagnostic deployment is performed automatically.
- Local original-video indexing, reviewed onset/settling and coordinate analysis, frame-sampling uncertainty, and annotated screenshots. Retimed or incomplete evidence requires review; no physical latency improvement is claimed.

Validation

- Diagnostic core compiled on macOS; finite-buffer/export and timestamp-analysis regressions passed. A synthetic 100 fps, ten-frame response measured 100 ms with a 90-110 ms sampling bracket.
- Actual-device instrumentation overhead, live gameplay, synchronized hand/screen recording, and accepted stationary-jitter baselines remain pending.

[0.3.2-rc.6] - 2026-09-06

Changes since rc.5. This candidate adds saved player setups, complete hand backups, signed controller sessions, and Setup/mobile refinements. Update both computers together: the default version-2 sender and receiver do not interoperate with older controller transport. The project base version remains 0.3.2; the release tag and installer manifests identify the candidate.

Added

- Up to twelve Controller-stored players with individual sensitivity, Academy progress, Glove Master awards, and saved calibration. Switching players pauses output and requires fresh centring or explicit same-position reuse.
- Complete version-2 hand-setup backups with name, personal and effective sensitivity, software identity, and per-player calibration. Restore reviews complete sensitivity and calibration reuse separately, preserves Academy progress, and recovers safely after interrupted writes. Earlier version-2 backups work; portable version-1 exports are rejected. Internal stores migrate to version 4 with private recovery backups.
- Version-2 HMAC-SHA256 controller messages with receiver-issued challenges, replay/retired-session rejection, bounded nonblocking handshakes, and restart recovery. No input state is queued during negotiation, and the shared secret is absent from signed packets.
- Idle matrix controls for On, Dim, or Off with faint app, console-service, authenticated-console, and independent Wi-Fi pixels. Setup reports unavailable Wi-Fi telemetry separately from disconnection. The unprivileged host sampler is installed on setup/upgrade; the fourth pixel requires the matching matrix firmware. Game, T/L, startup, error, and pairing displays are unchanged.
- Exact software commit/candidate metadata and running matrix firmware readback. Older firmware reports unavailable identity instead of an assumed match.

Changed

- Reorganized Setup wording and connection, pairing, attract, and power sections. One-time-code pairing is prominent; SSH password pairing remains available. Address checks distinguish name resolution from game delivery.
- Refined phone/tablet navigation, forms, and Academy/Play layouts down to 320 pixels.
- Moved hostname refresh off the controller send path without queues or new smoothing. Hardware microbenchmarks document packet processing cost separately from end-to-end latency.
- Extracted the shared page shell, Dashboard, Academy, Games, and tuning into maintained modules and removed obsolete UI definitions. Rendered pages remain byte-for-byte equivalent.
- Receiver defaults to signed input. The explicit temporary `--allow-legacy-controller` upgrade option closes after the first signed state and should be removed after migration.
- Updated README, Help, installation material, screenshots, architecture diagrams, PDF editions, and the maintained review parking lot.

Fixed

- Preserved the contacted Linux address/interface for UDP handshake replies when RetroPie has Ethernet and Wi-Fi on the same subnet. Replies require HMAC, fresh request/session identifiers, and the configured receiver port.
- Setup retries failed loads/actions, retains unsaved connection edits during refresh, and clears pairing approval after destination changes. Device-setting writes are serialized, private, and atomic; pairing-key changes disarm output.
- Start/Stop worker requests report pending delivery and retry only the latest intent. Supervised workers read pairing credentials from private configuration rather than process arguments.
- Receiver timeouts release native input and the virtual gamepad; rejected traffic and handshakes cannot postpone release. Malformed nested profile messages no longer terminate the listener.
- Glove Master replaces the final lesson panel after all sixteen lessons, avoiding an extra panel pushing the page downward. Start again restores the lesson panel.

Validation and remaining work

- All 355 tests pass on Python 3.7 and 3.12. Source/documentation audits, package verification, browser checks, and visual PDF inspection pass.
- Both devices ran the completed code at `a7131f8`. Cross-device dry-run tests accepted both fresh sessions, dropped negotiation frames, and rejected retired-session input. Production handshakes worked through both RetroPie addresses. Controller firmware matched, Wi-Fi status was connected, and saved hand settings, calibration, pairing, and Start/Stop choice were preserved.
- Full live gameplay, synchronized camera-to-display latency, stationary jitter, and fresh-device installation checks remain release gates. Earlier successful native Super Glove Ball gameplay does not substitute for repeating gameplay on this candidate. This is not the final 0.3.2 release.

[0.3.2-rc.5] - 2026-09-06

Cumulative release-candidate notes since 0.3.1. This candidate adds the refined idle matrix animation, reproducible live latency baselines, and updated native gameplay validation. Visible camera-to-game latency and a reliable stationary jitter baseline still need a coordinated video test; this is not the final 0.3.2 release. The project base version remains 0.3.2, with the candidate identity recorded by the release tag and installer manifests.

Added

- Added `scripts/measure-vision-status.py` for read-only, bounded latency baselines. It ignores duplicate cached status samples, separates changing run conditions, and reports observed timing distributions and optional neutral X/Y variation. Reports contain aggregate measurements, with explicit limits for unmeasured network, receiver, core, and display stages. Recognition and movement defaults are unchanged.
- Recorded separate live movement, requested-stationary, network round-trip, and native-publication cadence observations. The movement window retained detection in all 315 observed samples and measured 132.8 ms read-to-send p95; physical display latency and a reliable stationary-jitter baseline remain separate validation steps.
- Added a camera-controlled Rock Paper Scissors page at `/play`. A closed hand plays rock, an open palm plays paper, and the held V-sign plays scissors in a first-to-three match against Pixel Pal. It reuses the local practice-camera lease, pauses cabinet input while open, and provides touch and mouse controls. Help now points readers to its illustrated Gameplay Guide section, while This cabinet retains a direct link to the game.
- Confirmed every implemented native Super Glove Ball action during a completed live game: open-hand release/throw, fist grab/catch, index-point Robo-Bullet, and fist-plus-forward Power Punch, alongside Start and continuous X/Y. Movement is playable and substantially improved; residual latency remains the next refinement target. Unused wrist-rotation and packet-button fields stay neutral.

Changed

- Refined the idle matrix animation with a double-flash lightning bolt, separated fingers and thumb, a consistent cuff buckle, a longer fist hold, and a smaller travelling spark. Greyscale shading and a gradual glow finish the four-second loop. Added a preview rendered from the sketch's actual frames; physical LED appearance remains a separate check. Requires a matrix firmware update.
- Updated native compatibility test assertions to match the already documented completed-game confirmation and deliberately neutral unused packet fields.
- Corrected relay-test mock argument access for Python 3.7 compatibility.
- Adopted **VirtualGlove Controller** as the user-facing name for the Arduino UNO Q device throughout current guides, while retaining literal `uno-q` commands, filenames, host placeholders, and hardware-specific notes.
- Remembered the player's explicit Start/Stop controller choice across UNO Q application and system restarts. Registered RetroPie launches now maintain a renewable session only while RetroArch is running, allowing an armed controller to reconnect after an application restart without emitting into EmulationStation or the runcommand menu. Stop remains sticky; game exit, unknown games, and stale sessions release controls. Dashboard now distinguishes armed delivery from the live game session.
- Standardized numbered and bulleted lists across the Markdown guides. Built-in Help now keeps wrapped and loosely spaced items together, and generated PDFs keep every marker beside its complete item at page boundaries.
- Connected Super Glove Ball's native open hand, closed hand/fist, index-point, and calibrated signed-Z packet fields. The UNO Q publishes five-finger compound recognition explicitly; the custom Nestopia core emits `$00`, `$FF`, `$0F`, and absolute depth for throw, grab, Robo-Bullet, and fist-plus-forward Power Punch validation. Exact-ROM headless traces verify the ten-byte packet values; wrist rotation and remaining native button codes stay neutral.
- Replaced Glove Academy's threshold-first Tune panel with a Pixel Pal-guided personalisation wizard. Families choose the problem, confirm clear framing, record gesture-specific steps, pass a two-use preview test, and then save only affected shared components. Raw numerical controls now live under Advanced.
- Added dedicated **Show your hand** and **Find neutral** artwork, clearer Start and Select lesson titles, and an intentional **Set this as my centre** action.
- Added an optional local Academy diagnostic. It exercises the deployed proven tracker, emits an aggregate report without images or landmarks, and deletes its temporary video after analysis, cancellation, or a 30-minute abandonment.
- Added advisory hand-framing, darkness, and backlighting guidance without changing camera exposure automatically.
- Reduced the deliberate V-sign Start hold from 650 ms to 500 ms while retaining its short pulse, interruption cancellation, Menu Guard priority, and 300 ms visible-release rearm guard. Select remains a 150 ms hold.
- Evaluate directions, finger curls, wrist rolls, and Closed Hand on every fresh inference result. Glove Zap and Pull Back now reject stationary near/far hands and one-frame scale jumps by requiring two beyond-threshold observations plus 0.10 normalized palm-scale motion in the intended direction within 250 ms. Existing held, pulsed, turbo, and toggle mappings remain unchanged.
- Added rolling camera-read-to-send and changed-control-to-send p50/p95 readings to Dashboard, including continuous native X/Y changes. The native coordinate benchmark now verifies a deliberate step reaches 90% within 150 ms without adding another queue or core-side smoothing.

- Added local-only fixed and user-paced camera record/replay tools for identical-frame comparisons of MediaPipe Hands thread counts, experimental Tasks Video, 640×480 versus full-field 512×384, and preview-open versus preview-closed cost. The recorder covers near/far, gesture, jitter, depth, and recovery cues. Guided capture keeps a live browser preview visible, records each step only after player confirmation, and releases the camera when complete. Clips are temporary and never used for training.
- On the deployed UNO Q, preliminary 300-sample steady-state windows kept the 640×480 camera-read-to-decision p95 at 127.9 ms with Dashboard closed and 125.7 ms with its stream open. A 16-lane guided replay retained MediaPipe Hands at 640×480 with four threads: Tasks Video roughly doubled inference latency, and 512×384 did not meet the required 15% improvement while producing more false activations. Controller-enabled game measurements remain a separate acceptance step.
- Promoted the Project Overview PDF to the first position in built-in Help's Technical Documentation section and reordered that section from broad system orientation through increasingly specialized implementation material.
- Made the deployed legacy-lite MediaPipe tracker explicit while retaining an opt-in Tasks Video path for controlled comparisons. Camera capture now drains continuously into a newest-frame slot, background JPEG work can drop stale previews, and live diagnostics report frame freshness, inference cadence, skipped captures, and preview cost.
- Named the tracker choices **MediaPipe Hands** and **MediaPipe Tasks Video (experimental)** in Dashboard and documentation while retaining the stable `legacy` and `tasks-video` command identifiers.
- Replaced Glove Academy's mislabeled single-finger curl in the **Close your hand** lesson with a dedicated closed-fist illustration. The intentionally six-digit glove remains part of Pixel Pal's Extra-Digit Hunt, and the pose is now shown in the Gameplay Guide, Configuration Reference, quick reference, and Programs A-I manual wherever closed-hand recognition is taught.
- Made **Skip lesson**, **Previous**, and automatic lesson completion mutually consistent. Manual navigation now invalidates older recognition responses and pending automatic advances, so one button press changes the lesson once.
- Added guarded UNO Q recovery for one UVC camera. Installation may run without a camera; the first healthy sighting records the camera and its real parent hub in a root-owned allowlist, and a later healthy sighting updates the association if the camera moves. After a sustained outage the helper validates and resets only the last observed hub. Recovery is rate-limited, never loops during one outage, and leaves ordinary camera reopen attempts as the first response.

[0.3.1] - 2026-09-05

Documentation and Help maintenance release adding Pixel Pal's deliberately spoiler-free guide game and keeping packaged documentation verifiable.

Added

- Added Pixel Pal's Extra-Digit Hunt to the two illustrated gameplay guides. Readers count every preserved six-digit glove once per appearance, then find the verified per-guide answer at the back. Built-in Help keeps the answer behind an explicit reveal, accessible descriptions identify qualifying art, and a checked manifest prevents future artwork changes from making the answer stale. The App Lab package builder now refreshes its tracked checksum companion after each successful build so guide updates cannot leave a stale digest.

[0.3.0] - 2026-09-05

Stable feature release covering shared recognition, native Super Glove Ball support, simplified personal tuning, repeatable two-machine installation, and the completed illustrated documentation set.

Native Super Glove Ball and emulator support

- Confirmed how Super Glove Ball reads the original glove's ten-byte message, including detection, Start, and continuous horizontal and vertical movement. The byte values remain documented in the native compatibility guide.
- Corrected the custom Nestopia core so centre is exact and old movement is released when tracking, calibration, or the selected profile is invalid.
- Added reversible per-ROM native Nestopia/FCEUmm selection plus a RetroPie-only isolated core installer; stock Nestopia remains untouched.
- Audited all eight listed US ROMs: only Super Glove Ball consumes native multi-byte glove packets; the other games use standard controller-bit mappings through FCEUmm.
- Added a repeatable emulator comparison from the same saved game state. Native Super Glove Ball responded to movement within three displayed frames, while the FCEUmm Gun Smoke test responded within two.
- Compared the exact Super Glove Ball ROM in both cores, confirmed that FCEUmm remains standard-joypad-only, and corrected native Y wrapping so its packet and screen position span bottom, centre, and top.
- Added an optional RetroPie installer offer that builds the pinned GPLv2 native core locally, registers both per-ROM launch choices without changing the saved FCEUmm selection, and installs the upstream licence beside the core.

Installation and distribution

- Added versioned UNO Q and RetroPie installers with verified downloads, repeatable updates, backups, and read-only checks.
- Included the early-start and shutdown helpers in UNO Q installation; App Lab's command-line tools install the app and Arduino sketch.
- Added optional emulator installation and registered-game checks, including Bad Street Brawler's game-specific Glove Zap setting.
- Simplified installation instructions and moved manual repair and packaging details into the technical reference.
- Added release packaging and automated installer tests after fresh-device and physical gameplay validation on the UNO Q and RetroPie console.
- Added installation manifests shared by both package installers and Wi-Fi deployment. Updates back up and remove unchanged obsolete application files, preserve local edits, and provide interrupted-update recovery and read-only inventory checks.
- Allowed UNO Q deployments to select an explicit SSH identity and expanded the live deployment gate to verify every published technical guide and PDF.
- Made `config/profiles.json` release-owned and replaceable during updates so the shared recognition defaults supersede legacy per-game thresholds; the previous file is backed up and personal calibration and tuning under `data/` remain preserved.
- Hardened release validation so installer packages must contain the shared profile baseline and current recognition, tracking, tuning, and native-core selection modules. Update tests explicitly preserve neutral calibration and personal gesture tuning while replacing release-owned defaults.
- Preserved existing profile configuration during installation and Wi-Fi updates.
- Unified release/deployment payload selection and installer templates; removed unused helpers and state.

Recognition, performance, and safety

- Made hand centring accept only a complete, clearly detected hand. It averages several observations, measures natural resting movement, and replaces the previous centre only after the new sample is complete.
- Removed avoidable gameplay diagnostics: finger geometry is measured once, detailed landmarks are prepared only at the 5 fps preview cadence, and idle tuning skips measurement work while unchanged configurations are reused.
- Made Menu Guard and the V-sign mutually exclusive with a clear pinky deadband. Menu Guard now has priority, cancels pending Start pulses, and suppresses every ordinary controller button while the safety pose is active.
- Shared recognition settings across game profiles, reduced the hand travel needed for movement, and used measured resting jitter only when a player needs extra neutral space.
- Expanded Glove Academy from twelve to sixteen mapping-independent lessons with roll left/right, close hand, and a shared menu-guard pose. Practice polls every 75 ms and controller delivery remains paused.
- Hardened the V-sign Start command after live Gun Smoke/FCEUmm testing exposed accidental pause pulses. Start now requires a stable 0.65-second hold and a continuous 0.30-second non-V release before it can rearm; movement, A/B, and Select timing are unchanged.
- Reduced Start and Select pose debounce to 0.15 seconds before the later live-tested Start-specific hold and rearm protection, retaining one press per pose and release before repeating.
- Fixed controller session resets, malformed packet rejection, and timeout release during rejected network traffic.
- Applied finger release thresholds consistently in Programs E, F, and G.
- Matched tuning feedback to each recording step and made failed matrix commands retry.

Documentation and licensing

- Expanded the third-party runtime inventory for the modified Nestopia core with its patch checksum, affected files, runtime/install boundaries, GPL preservation rules, and pinned-update procedure. Documented stock RetroArch and FCEUmm as externally installed RetroPie dependencies and added an audit check that prevents the Nestopia revision or patch digest from drifting away from the published notices.
- Preserved Nestopia's original source copyright and GPL header verbatim, added a separate file-by-file VirtualGlove modification ledger, made the native-core build reject header changes, and installed that ledger beside the optional core and its upstream `COPYING` file.
- Centred gesture, profile, and matrix artwork in Help and printable table columns. Grouped related startup and attention-state matrix photographs into compact visual rows where side-by-side comparison is clearer.
- Added a dedicated menu-guard illustration and substantially enlarged the A/D/H experiment artwork in built-in Help and the gameplay PDF, with balanced text columns.
- Added the isolated `lr-nestopia-powerglove` research patch, reproducible pinned build, tracing, and guarded latest-sample bridge. Exact-ROM validation is now recorded alongside the retained FCEUmm fallback.

Added

- Added an optional permanent UNO Q early-start helper after a successful physical cold-boot trial. It releases the installed sketch earlier on each boot while preserving the system boot display and existing hourglass animation.
- Added a Matrix display guide in Help and PDF form, covering every display state, profile letters, pairing sequences, troubleshooting, and photographs of the physical display.
- Added a pulsing startup hourglass before Router Bridge initialization. A dedicated Arduino sketch display task keeps it moving while Linux is starting; ordinary app status then selects the glove, profile, or Academy display. The protected system boot display remains unchanged.
- Renamed the Learn section to Glove Academy in navigation, the page heading, and current guides. Kept `/learn` links, lesson/tuning behaviour, and L/T matrix indicators unchanged.
- Added startup stage timings for library imports, model preparation, camera initialization, and first inference.
- Added individual gesture illustrations and Pixel Pal to the website and friendly manuals, with a smaller PNG for web use.
- Preserved and bundled the unmodified Google Hand Landmarker model with Apache 2.0 licence text, provenance, and checksum.
- Added offline model-cache installation and recovery from the bundled copy, with verified download fallback only when the bundle is absent.
- Made package builds verify the model and required licence files before reporting success.

Changed

- Preloaded OpenCV and MediaPipe in the background when the vision worker starts, keeping controls responsive and the camera off until requested. On the tested cabinet, first activation after a reboot took 1.21 seconds once preloading completed.
- Made `/dashboard` the default page, retaining redirects from `/` and `/debug`.
- Made web footers use the release version, with a development indicator on dev builds.

Gesture tuning simplification

- Replaced seven recordings with open hand, gesture, open hand (three seconds each).
- Added optional all-finger hand setup using a gentle fist with the thumb outside.
- Added live extended/curled finger feedback sharing V-sign and thumbs-up recognition checks. Existing saved thresholds remain compatible.
- Completed live camera validation on the installed UNO Q system.
- Featured common tuning controls and grouped directions, wrist rolls, and other combinations under More adjustments. Added explicit movement-specific starting-position and return instructions for Glove Zap, pull-back, directions, and wrist rolls.
- Added pull-back as the twelfth ordinary Learn lesson, with a live action indicator, backward-distance feedback, and personal pull thresholds.
- Gameplay movement mappings now use shared activation/release states for wrist controls, forward push, pull-back, movement answers, and braking. Preserved game button mappings, pulses, and pull-toggle edges; clear depth activation on tracking loss.
- Matched Tune's T animation to Learn's L using the same scan line, trailing glow, brightness, and frame timing.
- Standardized Glove Zap and Pull Back labels across Learn, Tune, and the guides.

Documentation and firmware reconciliation

- Documented optional five-finger hand setup, three recordings of three seconds, preview/save/discard/reset, extended-only finger handling, and unchanged version-1 saved settings.
- Updated ordinary Learn to twelve lessons with Glove Zap and Pull Back, and documented shared gameplay activation/release states while retaining profile mappings and menu holds.
- Documented matching scanning L/T animations and consistent action names.
- Recorded the verified Zephyr 1.0.0 platform and pinned sketch libraries, distinguishing platform installation, compile-only validation, and firmware upload.
- Refreshed maintained guides and their PDF editions. Historical entries above retain the behaviour and names at the time of each change.
- Completed live camera, game-control, cold-boot, and physical display checks on the installed systems in addition to the automated release checks.

Complete-pose validation before tuning suggestions

- Automatic analysis now requires all selected fingers to match together in at least 90% of accepted samples, including opening and final release phases.
- V-sign validates extended index/middle alongside curled ring/pinky; thumbs-up validates extended thumb alongside four curled fingers, using shared recognition checks and existing personal extension thresholds.
- Failed analysis identifies the finger/phase and clears the preview without modifying saved settings. Existing holds and orientation-flexible thumb extension remain unchanged.
- Added regressions for invalid extended fingers in each phase, personal thresholds, exact boundaries, noise tolerance, simultaneous pose failures, invalid samples, and saved-value preservation, then completed live camera validation.

Bad Street Brawler Glove Zap

- Mapped forward-push activation to a 180 ms simultaneous Left + Right pulse in the Bad Street Brawler profile. Require release before retriggering; cancel on menu poses, tracking loss, and recalibration.
- Enabled opposing directions in Bad Street Brawler's FCEUmm game options while preserving global settings. The existing RetroPie receiver and gamepad mapping need no changes.
- Corrected the earlier documentation claim that this attack required native glove support. Added pulse, cancellation, rearming, and other-profile regressions; all 155 tests passed. A two-frame headless test confirmed the game-specific options load. Live attack validation remains outstanding.

Repeatable RetroPie Glove Zap configuration

- Added `powerglove-bsb-zap` with read-only checks and explicit `--apply` setup.
- Checks installed FCEUmm/RetroArch and game emulator selection; reports missing dependencies or incompatible selection without silently changing either.
- Preserves inherited options and global settings; backs up and atomically updates the game file, with repeat-run detection and rollback instructions.
- Added six isolated tests for creation, preservation, inheritance, missing cores, wrong emulator selection, read-only mode, running games, redirects, and symlinks.
- Documented installation, use, limitations, and the remaining live gameplay check.

[0.2.5] - 2026-09-04

This release adds web game mappings, personal gesture tuning, reliable game-launch profile selection, and refreshed illustrated manuals.

Added

- Added a dedicated T on the UNO Q matrix while gesture tuning is active.
- Added a Games JSON editor with paired RetroPie access, duplicate-name validation, conflict detection, verified saves, backup download, and restoration.
- Added guided Learn tuning with camera measurements, independent gesture thresholds, temporary previews, and persistent personal adjustments shared across profiles.
- Added a paired-console Games service so Setup can safely read and update the RetroPie game list. Installation and health checks verify it automatically.

Fixed

- Moved game mapping editing into Setup and compacted Learn tuning, placing threshold values beneath the camera.
- Kept game-profile commands available even while the camera is starting, and improved error messages when a launch request or configuration is rejected.
- Kept profile changes responsive during blocked camera startup or reads, and reused the camera and tracker when switching between active profiles.
- Added exact compressed ROM filenames to the default registry so supported `.zip` and `.7z` games can select their profiles.

Documentation

- Refreshed all six interface screenshots on September 4, 2026, blurring camera imagery before capture.
- Illustrated Setup's Games section and Learn's compact Tune layout across the guides, including the T matrix indicator.
- Regenerated the printable manuals to match the updated Markdown and screenshots.
- Reorganized the overview and installation guide around a complete, numbered setup path: Git download, App Lab package import, UNO Q host setup, RetroPie installation, pairing, and gameplay checks.
- Added Programs A-I controls to the overview and a complete command-line reference with project flags, defaults, arguments, and the system-command options used in the guides.
- Applied reader-focused writing conventions across the guides, added first-round game exercises, and corrected the RetroPie update path. PDF regeneration remains a separate publication step.
- Reviewed all Markdown guides for natural English, replacing sentence fragments and compressed notes with complete explanations while retaining concise tables and release entries.
- Revised the Quick Reference with complete installation prerequisites, camera inspection commands, embedded screenshots, calibration explanations, and step-by-step game registration. Clarified shutdown readiness and local tests.
- Corrected stale lesson counts, profile descriptions, calibration behaviour, pairing guidance, reboot verification status, and receiver removal steps.
- Proofread the Quick Reference and normalized Markdown list indentation and continuation text for consistent rendering.
- Regenerated all ten PDF editions, repaired internal section links, and improved heading and image pagination.

[0.2.0] - 2026-09-03

Added

- Added reliable `.local` hostname lookup that survives App Lab rebuilding the application containers.
- Added one-command host setup for RetroPie and UNO Q, with managed-file backups, preserved private configuration, delayed startup, and read-only PASS/FAIL/ACTION checks. Empty pairing tokens no longer cause restart loops.
- Added explicit A, B, and GLOVE ZAP practice lessons and live indicators. Learn consistently uses the general profile without changing the selected game.
- Added a Glove Master completion achievement to Learn after all eleven lessons are recognised, with Start again and Dashboard actions. Skips do not count.
- Added hand illustrations to every Learn lesson and finger/pose feedback for Start and Select. Learn accepts confirmed menu poses after their short button pulse ends, rather than requiring a second hold longer than the pulse.
- Added a live active-profile selector to the Dashboard without changing the startup profile saved on Setup.
- Added temporary Learn sessions that start vision while preserving the selected profile and desired controller state, including multi-tab leases and automatic recovery when a page disappears unexpectedly.
- Added a dedicated matrix gestures-idle state with a pinball-style animated glove, separate from both true shutdown and the flashing error X.
- Added a dedicated Learn-mode matrix state with a bright `L` and moving greyscale scan highlight.
- Restored safe-shutdown readiness automatically after a reboot or App Lab application replacement.

Changed

- Shutdown now requests a graceful system halt instead of poweroff, which was observed to reboot the UNO Q. Reinstall the host helper to apply this change; hardware tests subsequently confirmed automatic restart with both the powered hub and direct Mac USB connection. Remaining halted is a known unresolved limitation.
- New installations leave the RetroPie destination blank, show generic hostname examples, and keep local practice available before pairing. Controller start requires a destination; saved destinations survive updates.
- Added persistent `.local` hostname lookup for gameplay and pairing, with a short address refresh when a device moves on the network.
- Standardized Dashboard/Learn Calibrate actions with red busy and blue completed feedback, consistent navigation buttons, and shorter Connection/Shutdown labels.
- Prioritized controller transmission before matrix updates and limited browser JPEG encoding to 15 fps; added inference_ms and send_ms diagnostics.
- Prepared SSH pairing dependencies separately in the persistent runtime cache so dependency downloads do not consume the SSH connection deadline.
- Tuned thumbs-up/Select closed-finger detection to 0.42 using live pose measurements, retaining the straight-thumb requirement and deliberate hold.
- Tuned V/Start closed ring and pinky detection to 0.42 using live pose measurements, retaining straight index/middle checks and the deliberate hold.
- Tuned curl activation/release to 0.50/0.35 using live comfortable index-curl measurements. Learn now follows the same hysteresis state as gameplay, independent of pulsed buttons. Calibration resets held finger switches.

- Finger curl now uses the strongest joint bend and includes the base knuckle for the four fingers. Learn exposes exact curls, a magnified landmark view, and forward-movement readings. Push lessons use continuous feedback so a one-frame event cannot be missed by browser polling.
- Camera finger curl now uses 3D world landmarks, with an aspect-corrected normalized-depth fallback. Movement and gesture thresholds are unchanged.
- Made Dashboard and Learn show camera/tracker startup with elapsed time and a first-start explanation, without enabling the camera in Gestures off.
- Kept Learn startup feedback updating before camera frames arrive and disabled centring until vision is active.
- Replaced generic Program A-I labels on Dashboard and Setup with the program letter and its intended game or use, while retaining the existing profile IDs.
- Standardized the reader-facing game name "Gun Smoke" throughout Help and the public guides; exact `Gun.Smoke` ROM basenames remain unchanged for matching.
- Made leaving Learn restore the selected profile, camera state, and controller state. Loading or refreshing the Dashboard now clears abandoned Learn sessions, prevents their old heartbeats from reactivating vision, and reapplies the selected mode.
- Made Wi-Fi deployments preserve VirtualGlove as the UNO Q default startup app so the dashboard returns after a board reboot.
- Made Wi-Fi deployments restore the shutdown readiness marker when the installed host watcher is active.
- Extended deployment health verification to tolerate a three-minute cold App Lab runtime startup.
- Made deployment verification use the UNO Q address from the active SSH connection instead of accidentally selecting a Docker bridge interface.
- Made **Gestures off** a healthy worker state that releases controller input, closes the camera and MediaPipe tracker, and keeps the website and authenticated RetroPie profile listener available.
- Made camera and model initialization lazy so idle mode performs no capture or vision processing and can return to an active profile without restarting the website.
- Reworked the matrix attract sequence into distinct pinball-style beats: a four-frame energy sweep, a broad travelling cuff, a staged glove reveal, intermediate finger curls, an eight-position spark with a comet trail, one outline pulse, and a readable hold.
- Used the UNO Q matrix's full eight-level greyscale range to separate the dim glove body, spark halo, bright spark, and whole-glove pulse.

Documentation

- Documented the UNO Q matrix as an eight-level monochrome DMD/BitPixel-style design target, including silhouette, contrast, motion, pulse, and physical review guidance for future animations.
- Standardized source headers with each file's purpose, author, copyright, SPDX licence identifier, local history, and links to the complete history.
- Documented public interfaces and non-obvious security, lifecycle, tracking, packaging, and rendering functions.
- Added this centralized project changelog and its print-ready PDF edition.
- Added an automated audit for required source headers, module descriptions, and production Python interface docstrings.
- Added a complete configuration reference covering JSON templates and active copies, gesture thresholds, manifests, RetroArch mapping, systemd units, generated state, secret handling, and the App Lab installation ZIP location.
- Added GitHub Actions checks for supported Python versions, tests, source and documentation audits, syntax, PDF builds, and App Lab installation ZIP verification.
- Added a security policy for private reporting, pairing and network boundaries, shared-token handling, shutdown permissions, and dependency integrity.
- Added a contributing guide for code style, tests, documentation, changelog, generated artifacts, commits, and pull requests.
- Added reusable documentation and App Lab installation package verification tools.
- Added an illustrated, one-page-per-game handbook for all eight automatically configured titles, with original direction, finger-pose, wrist, and depth art.
- Placed cropped gesture illustrations beside the corresponding instructions in the gameplay handbook, including the universal V-sign and thumbs-up controls.
- Added the same contextual gesture illustrations to every Program A-I card and documented the menu gestures shared by all nine programs.
- Added an off-script gameplay section that encourages safe experiments with other NES and Famicom games, especially the unassigned A, D, and H programs.
- Updated the cabinet cheat sheet with off-script profile testing, safe behaviour for unknown games, and exact ROM registration guidance.
- Renamed the built-in installation Help route, retained the original URL as an alias, and aligned its summaries with the Play Checklist terminology.
- Extended the documentation audit to require Help-library coverage and a gameplay section for every title in the shipped game registry.
- Expanded Wi-Fi deployment verification to confirm the current installation and gameplay guides, raw Markdown, and gesture artwork on the UNO Q.
- Fixed the Help renderer so the allowlisted gesture images embedded in control tables appear alongside their actions, while unsafe raw HTML remains escaped.
- Extended UNO Q deployment checks across every Help guide and representative gameplay and Programs A-I illustrations.

- Documented the required Markdown-to-PDF workflow and UNO Q Help synchronization steps for documentation contributions.
- Added offline PDF links to every public Help guide and the project overview, while keeping the cabinet-specific quick-reference PDF private.
- Included the nine allowlisted public PDFs in UNO Q deployments and App Lab installation packages, with package and live-route verification.
- Established `dev` as the integration branch, documented release and hotfix promotion into `main`, and enabled CI validation for pushes to both branches.
- Refreshed the Dashboard, Learn, and Setup screenshots from the running UNO Q after the tagline, compact diagnostics, and safe-shutdown controls were added.
- Added an offline Help library that renders the maintained public Markdown guides with responsive navigation, contents links, illustrations, tables, code samples, and access to the original source.
- Added a dynamic **This cabinet** Help page whose UNO Q links follow the browser's validated hostname or IP and whose non-secret RetroPie values come from the active configuration.
- Rewrote the configuration reference for public installations, with guided UNO Q and RetroPie setup, field-level behaviour, safe gesture tuning, network boundaries, backup and recovery advice, and symptom-based troubleshooting.
- Renamed the Field Guide as the Installation Guide and generalized camera instructions for UVC-compatible USB cameras while recording the Razer Kiyo as tested reference hardware.

Fixed

- Changed Wi-Fi deployment to use SFTP staging and terminal-backed remote commands for UNO Q systems that stall non-terminal SSH sessions.
- Allowed the UNO Q deployment health check to use the board's current IP when its `.local` name pauses during a container restart.
- Made shutdown requests appear as one complete action, preventing the host helper from seeing a half-written request.
- Updated the GitHub Actions workflow to use the current Node 24 action releases and run the Python 3.7 compatibility job on Ubuntu 22.04.
- Corrected Program I so index curl accelerates in Knight Rider, a forward push accelerates with turbo, and thumb curl fires the weapons.

0.1.0 - 2026-09-03

Added

- Camera-only Power Glove tracking on Arduino UNO Q with MediaPipe.
- Gesture profiles for Bad Street Brawler, Super Glove Ball, and cartridge-free Programs A-I.
- Added authenticated game-profile selection and a virtual Linux gamepad for RetroPie, selected automatically when a registered game launches.
- Dashboard, live diagnostics, offline gesture lessons, configuration controls, camera recovery, controller start/stop controls, and UNO Q matrix feedback.
- Wi-Fi deployment, App Lab packaging, runtime-asset retrieval, branded PDF generation, and a fixed-purpose host shutdown helper.
- Project overview, installation guide, quick reference, profile handbook, third-party component notice, screenshots, and reproducible build instructions.

Changed

- Delayed RetroPie receiver startup until EmulationStation finishes its initial device scan, preventing the virtual controller from disrupting cabinet input.
- Made the controller start disarmed and kept tracking/dashboard operation alive when RetroPie or the camera temporarily becomes unavailable.
- Moved private device settings and downloaded model data outside the App Lab installation ZIP.
- Changed the App Lab installation ZIP to download Google's Hand Landmarker model on first launch and verify its pinned SHA-256 digest before installation.

Fixed

- Recovered cleanly from USB camera disconnects and slow UVC camera wake-up.
- Kept hand tracking responsive when the console, Wi-Fi, or local hostname lookup temporarily disappears.
- Corrected password pairing so credentials never appear in command arguments and the shared token is transferred and installed reliably.
- Corrected UNO Q dependency isolation, secure token upload, PDF builder file mode, landscape diagnostics layout, and cabinet launch integration.

Security

- Added encrypted pairing with certificate comparison, a short-lived physical PIN, time-limited connection attempts, and private pairing files.
- Required confirmation and a fixed host-side request path for system shutdown.
- Added a third-party component notice covering licences, provenance, pinned versions, checksums, and update procedure.

Neutral calibration retention

The app now saves completed calibration so the reference survives transitions between Learn and gameplay, profile changes, camera reconnections, and restarts. When you recalibrate, the app replaces the saved reference atomically.

RetroPie mDNS installation

The RetroPie and UNO Q installers now provide the system components needed for `.local` addresses such as `virtualglove.local`. Existing hostnames and pairing settings are preserved. At the time of this entry, a completely fresh RetroPie installation had not yet been tested.

Documentation and shutdown wording consistency

The documentation now explains when to recalibrate, which files to back up, and what the hand-identification score means. Shutdown confirmations also explain that the UNO Q may restart automatically.